Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kilifi.

Kilifi County Climate Change Adaptation Action Plan (2025–2030)

Consultant

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1 Chapter 1: Background information

1.1 Introduction

Plan International Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) project in Kilifi and Kwale Counties, two coastal counties endowed with ecologically and economically vital marine and terrestrial ecosystems. In Kilifi county, these ecosystems include mangrove forests, coastal woodlands, coral reefs, seagrass beds and extensive marine waters that support critical livelihood sectors such as small scale fisheries, tourism, maritime trade and agriculture. These ecosystems provide vital services including shoreline stabilization, biodiversity conservation, carbon sequestration and nutrient cycling, forming the ecological and economic foundation of the county.

Kilifi County covers approximately 12,609 km² with a population of 1,453,787 persons (704,089 males and 749,698 females) according to the Kenya National Bureau of Statistics (KNBS, 2019) (Figure 1). The county experiences a tropical climate with bimodal rainfall patterns, receiving long rains between March and May and short rains between October and December. Average annual temperatures range between 23°C and 31°C, with the coastal belt experiencing higher humidity and temperatures compared to the inland plateau. The county's economy is predominantly driven by climate-sensitive sectors such as rain-fed agriculture, fisheries, livestock, and tourism, making it highly vulnerable to the effects of climate variability and change.

According to the Kilifi County Climate Risk Profile 2021, the major climatic hazards affecting the hazards affecting the county include recurrent droughts, flash floods, sea level rise, salt water intrusion, coastal erosion and degradation of forests and mangroves. These climate induced hazards have resulted in reduced agricultural activity, loss of fisheries habitats and increased poverty levels, disproportionately affecting vulnerable groups including women, youth, and Persons Abled Differently (PAD).

To address these challenges, the COSME project applies an ecosystem-based and gender-responsive adaptation approach to strengthen community resilience and promote sustainable livelihoods. The initiative supports the formulation of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) through participatory climate risk assessments that integrate local knowledge with scientific evidence. These ward-level adaptation and preparedness plans form the basis for developing the Kilifi County Climate Change Adaptation Action Plan (CCAAP 2025–2030), ensuring that county strategies are grounded in localized realities, participatory governance, and gender equity.

In Kilifi County, the wards of Mwarakaya, Mnarani, and Sokoke were selected for assessment to represent the county's diverse coastal ecological systems and socio-economic conditions. These wards were prioritized due to their proximity to sensitive marine and forest ecosystems, dependence on natural resources for livelihoods, and exposure to multiple climate hazards.

The findings from these wards reveal localized vulnerabilities and adaptive needs that highlight the urgency of coordinated climate action at both ward and county levels. Consequently, the Kilifi County Climate Change Adaptation Action Plan consolidates insights from the ward-level processes and aligns them with key national frameworks such as the Climate Change Act (2023), the National Climate Change Action Plan (NCCAP), and the National Adaptation Plan (NAP). This alignment ensures consistency with Kenya's devolved climate governance architecture while promoting inclusivity, equity, and sustainability in climate adaptation planning and implementation.

This plan aims to mainstream climate resilience into all sectors, safeguard ecosystem integrity and build adaptive capacities of communities, particularly women and other vulnerable groups to respond effectively to the increasing impacts of climate change.

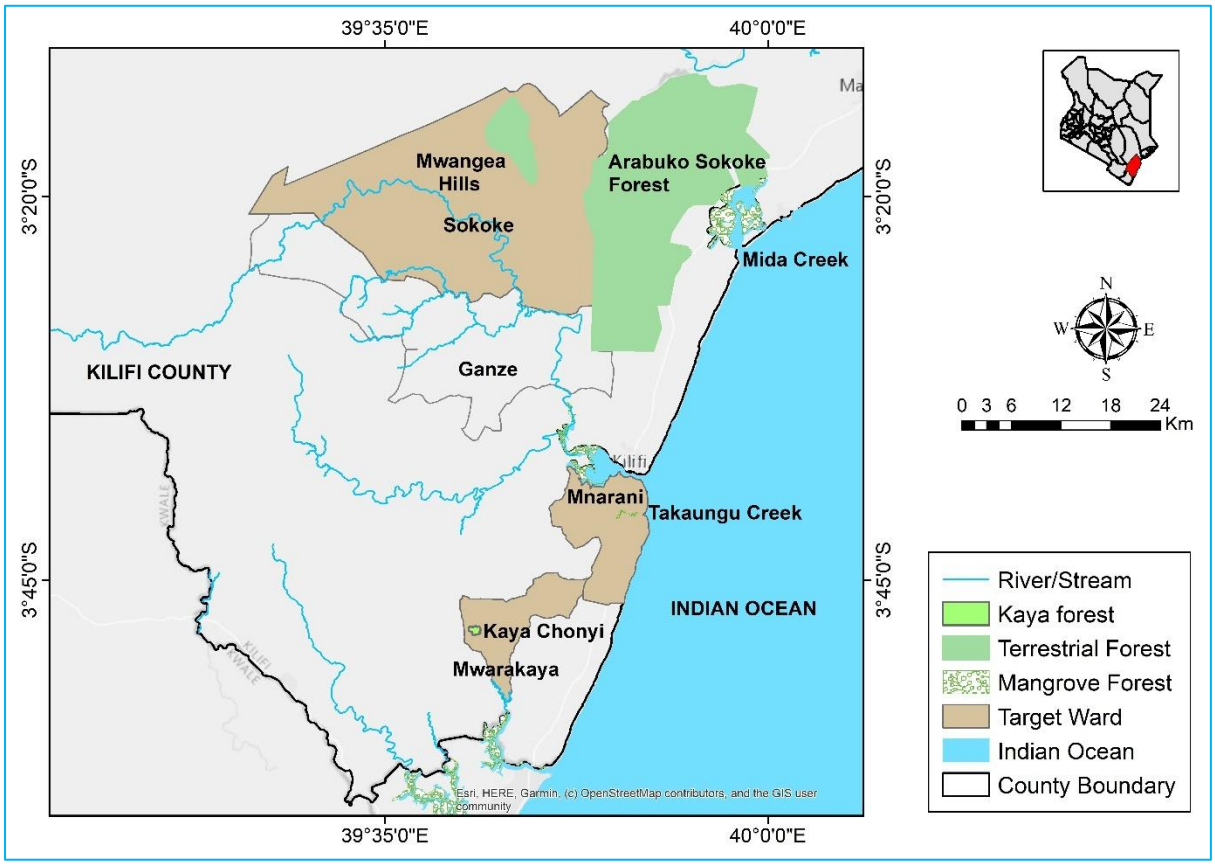

Figure 1: Location of Kilifi County and survey sites (Source: PCVA,2025)

1.2 Plan development approach

The Kilifi County Climate Change Adaptation Action Plan (CCAAP 2025–2030) was developed through a participatory, evidence-based, and gender-responsive process that builds upon the foundations laid by the Ward Adaptation Plans (WAPs) for Mnarani, Mwarakaya and Sokoke Wards. The plan draws extensively from Participatory Climate Vulnerability and Risk Assessments (PCVAs) conducted across fourteen (14) villages, namely Takaungu, Mkwajuni-Takaungu, Vuma, Vijiweni, Midzimitsano and Timboni in Mnarani Ward: Mbuyuni, Ondoni, Vwevwesi, Silala in Mwarakaya Ward and Dikunze, Ndungumnani, Mwele, Midugani in Sokoke Ward. These sites were selected to represent the ecological, socio-economic, and cultural diversity of Kilifi County's coastal landscape, while also reflecting areas of high climate exposure and ecological sensitivity within the COSME Project's intervention scope.

The development of the CCAAP was anchored in a livelihoods-based vulnerability framework that recognizes vulnerability as the outcome of interrelated factors i.e., exposure, sensitivity, and adaptive capacity, shaped by ecological conditions, social networks, economic assets, and institutional systems. This framework underscores that vulnerability is experienced differently across population groups, with women, men, youth, and Persons Abled Differently (PAD) facing unique climate risks and adaptation barriers. By adopting this gender-responsive perspective, the CCAAP ensures that the identified adaptation measures address the differentiated needs, capacities, and priorities of diverse social groups within the county.

The development of the CCAAP was anchored in a livelihoods-based vulnerability framework that recognizes vulnerability as the outcome of interrelated factors i.e., exposure, sensitivity, and adaptive capacity, shaped by ecological conditions, social networks, economic assets, and institutional systems (Figure 2). This framework underscores that vulnerability is experienced differently across population groups, with women, men, youth, and Persons Abled Differently (PAD) facing unique climate risks and adaptation barriers. By adopting this gender-responsive perspective, the CCAAP ensures that the identified adaptation measures address the differentiated needs, capacities, and priorities of diverse social groups within the county.

The CCAP was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, artisanal fishing, seaweed farming, mangrove harvesting, small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

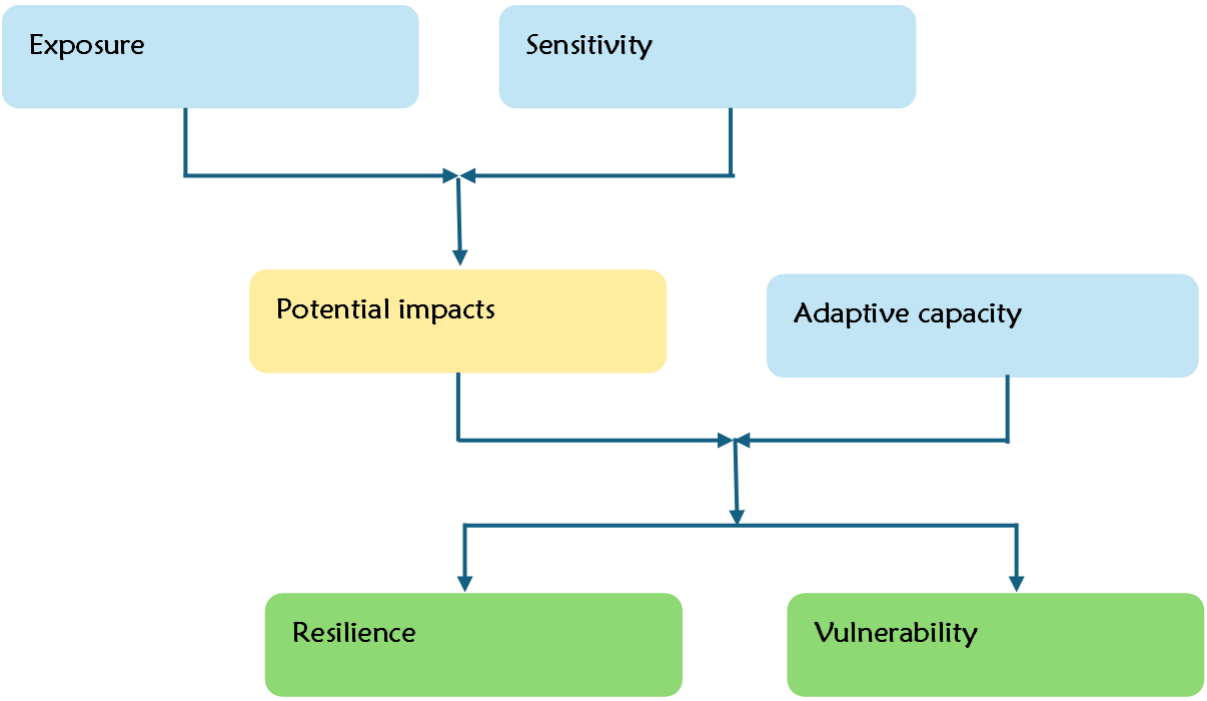

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014).

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kilifi County was reviewed, data gaps specific to the three wards were identified, and a sequenced work plan was developed outlining participatory diagnostics,

household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within the county, including the County Climate Change Planning Committee, ward administrators, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, seaweed farming groups, mangrove user groups, BMUs, WRUAs and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The CCAAP was informed by a comprehensive PCVA process implemented in 14 COSME target villages within Mnarani, Mwarakaya and Sokoke wards. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the CCAAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the county was delineated using GIS shapefiles provided by Plan International and Kilifi County Government. Within this boundary, the 14 COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, susceptibility to strong winds and sea-water intrusion, and documented human–wildlife conflict incidents. Village profiles were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

Ward-level core participatory analysis teams were established for Mnarani, Mwarakaya and Sokoke ward drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture, livestock and fisheries were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection.

Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across the three wards. This phase synthesized literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the 14 COSME villages, and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working Group I and II, particularly projections relevant to East African coastal systems, including accelerated warming rates, sea surface temperature rise, intensification of short-duration extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

At the regional scale, the review integrated IGAD Climate Prediction and Applications Centre (ICPAC) bulletins, the Africa Climate Policy Centre (ACPC) projections for Eastern Africa, and Western Indian Ocean Marine Science Association (WIOMSA) research on mangrove decline, seaweed vulnerability, fishery collapses and oceanographic shifts. This regional evidence positioned Kilifi within a broader coastal climate risk corridor characterized by rising temperatures, altered monsoon circulation, changing upwelling systems and degradation of nearshore ecosystems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kilifi County Integrated Development Plan (CIDP III, 2023–2027), the Kilifi County Climate Change Act (2021), the County Climate Change Fund (CCCF) Regulations, the County Spatial Plan, and sector strategies in agriculture, livestock, fisheries, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), marine ecosystem studies from KMFRI, forestry and

mangrove assessments by the Kenya Forest Service (KFS), fishery catch records from the State Department for Fisheries, and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from the wards could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the 14 COSME villages to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures, shifts in wind patterns and increased intensity of episodic storms and coastal surges. Participants described the material impacts of these climatic variations on agricultural productivity, seaweed quality and harvest cycles, artisanal fisheries (including reduced catch per unit effort), mangrove health, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced fishing days caused by strong winds, damaged boats during storms and increased anxiety over household provisioning responsibilities. In addition, participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies. These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: Focus Group Discussions in Sokoke Ward (Source: Field Surveys, 2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the 14 villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, irrigated horticultural fields, mangrove stands and adjacent intertidal zones, seaweed farming lines, fish landing sites, riparian zones, boreholes and shallow wells, public institutions (schools, health dispensaries), access roads, market areas, and known flood plains.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder roads due to repeated flooding, heavily eroded gullies formed by runoff from bare catchments, saline intrusion into agricultural fields and domestic wells, siltation of wetlands, collapsed seaweed lines following strong winds, degraded mangrove stands exhibiting canopy thinning or sediment stress, and infrastructure damage such as washed-out culverts and submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure.

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape. Through participatory sketching, communities identified settlement clusters, farmlands, grazing areas, mangrove forests, seaweed farms, rivers, swamps, markets, schools, health facilities, water points and religious centres. Hazard features; such as recurrent flood zones, storm-surge-prone coastal segments, areas with saline intrusion, strong-wind corridors, were marked and discussed. This mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, storm events, pest invasions (armyworms, whiteflies and rice pests), fishery fluctuations linked to monsoon variability, severe seaweed farm losses from temperature anomalies and episodes of heightened human–wildlife conflict. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, tidal patterns, agricultural operations, fishing seasons, seaweed planting and harvesting, mangrove harvesting, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-season rainfall, persistent dry spells and shifts in traditional fishing and seaweed farming seasons. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as farming, seaweed farming, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to weather-inhibited fishing, and youth described

increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, fishing activity, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as farming, fishing, seaweed sales, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, strong winds disrupted fishing and seaweed farming, leading to economic losses, debt accumulation and psychological stress. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.4 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in the three wards. Hazards such as drought, floods, environmental degradation, strong winds, sea-level rise, human–wildlife conflict and agricultural pests and diseases were arranged against critical assets including rain-fed farms, irrigated plots, mangrove forests, seaweed farms, artisanal fisheries, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the high susceptibility of mangrove areas and seaweed farms to combined effects of temperature stress, strong winds and changing salinity regimes; the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways.

1.2.2.5 Step 5: Analysis of Community Action (Tool 9 – Venn Diagrams)

Analysis of existing coping and adaptation practices revealed a range of coping strategies and externally supported interventions. The Venn diagram exercise (Tool 9) enabled communities to map the institutional ecosystem influencing adaptation in the ward. Institutions such as county line departments, NGOs, community-based organisations, religious institutions, savings and credit groups, fisheries and forestry officers, COSME project structures, and private sector actors were placed on diagrams according to their perceived influence and accessibility. The spatial arrangement revealed a significant presence of institutions but varying degrees of functional support. While some organisations maintained close, supportive relationships, such as community groups collaborating with NbS and seaweed farmer associations, others operated at a distance, limiting their impact. This analysis also identified critical gaps, including insufficient extension services for seaweed and fisheries, weak linkages to financial services, limited climate information dissemination, and inadequate institutional frameworks for systematic PAD inclusion. These findings provided essential insights for designing governance and institutional strengthening elements within the County Adaptation Plan.

1.2.2.6 Step 6: Drawing Action from Analysis (Tool 10 – Adaptation Pathways)

The final step in the participatory diagnostic process involved translating identified vulnerabilities, exposure pathways and institutional gaps into a coherent adaptation strategy through the Adaptation Pathways tool (Tool 10). Communities revisited impact chains and vulnerability dynamics to identify feasible response strategies across short-, medium- and long-term horizons. The exercise involved structured deliberations that considered gendered labour implications, the availability of financial and natural resources, household and community capacities, and alignment with ongoing COSME and county initiatives.

Resulting adaptation pathways articulated sequenced actions for key socio-ecological systems; agriculture, water resources, mangrove and terrestrial forests, seaweed farming and fisheries, community infrastructure, early warning systems and social protection mechanisms. This included immediate resilience-building measures such as water harvesting, reinforcement of seaweed lines, community drainage clearance and household-level livelihood diversification; medium-term actions such as improved coastal protection, establishment of climate-resilient seaweed strains, expansion of irrigation infrastructure and strengthening local governance structures; and long-term interventions such as large-scale mangrove rehabilitation, institutionalization of community-based climate monitoring systems, and improvement of rural infrastructure. These pathways formed the backbone of the County Adaptation Plan, linking diagnostic insights to strategic action.

1.2.3 Household Survey

A structured household survey was administered to 295 households in the study area to provide quantitative evidence complementing qualitative findings. Sampling was designed to include households across diverse livelihood portfolios and different vulnerability categories, including female-headed households, large households with high dependency ratios, households with PAD members, and those primarily reliant on climate-sensitive livelihoods such as rain-fed agriculture, artisanal fishing and seaweed farming. Primary respondents were predominantly women (approximately 65 %), ensuring robust capture of intra-household dynamics, care burdens, access to services and lived experiences of climate stress.

The survey collected data on demographic structure, education, disability status, land tenure and resource access, livelihood contributions, income diversification, exposure to recent climatic shocks, losses and damages, coping strategies, food security indicators, institutional linkages and participation in social protection and savings groups. Data were collected digitally using KoboToolbox to ensure consistency and reduce transcription errors. These data provided a

statistically grounded basis for validating PCVA findings and understanding patterns of vulnerability at household level across gender, age and livelihood categories.

1.2.4 Current Mangrove and Forest Health Assessment

To assess the condition of key coastal and terrestrial ecosystems supporting nature-based adaptation in the county, a remote sensing analysis was conducted using Sentinel-2 imagery at 10-metre spatial resolution. Imagery underwent standard pre-processing steps including radiometric correction and cloud masking before the Normalized Difference Vegetation Index (NDVI) was computed using the formula.

$$\text{NDVI} = \frac{NIR - RED}{NIR + RED} \qquad \text{Equation 1}$$

Where RED is visible red reflectance, and NIR is near infrared reflectance.

NDVI values were categorized into low, moderate and high vegetation vigor classes. Low NDVI signatures corresponded to degraded or stressed vegetation, often reflecting factors such as saltwater intrusion, wood extraction pressure, sedimentation and hydrological alteration. High NDVI zones served as ecological benchmarks, indicating healthier vegetation stands. These spatial patterns were subsequently compared with PCVA hazard–resource maps and field observations to identify priority sites for ecological restoration, mangrove replanting, mangrove hydrological rehabilitation and sustainable forest management within the ward.

1.2.5 Current and Future Climate Projections

Climate projections relevant to the county were developed supplemented by remote sensing and historical observational datasets. Daily rainfall Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Actual Evapotranspiration (AET), Potential Evapotranspiration (PET), (from ERA5-Ag) and temperature variables (minimum, mean, maximum and Land Surface Temperature (LST) from Landsat and ERA5-Ag) were extracted through Google Earth Engine after spatial boundaries had been validated in QGIS. The resulting datasets were aggregated into annual totals or means and subsequently analysed using Python (Colab) employing pandas, numpy, scikit-learn and matplotlib.

Linear regression models were fitted to each climatic variable to estimate historical trends and project future changes to 2035 and 2045. Temperature projections incorporated Intergovernmental Panel on Climate Change (IPCC) AR6-aligned warming increments for East Africa's coastal belt, applying approximately +0.02°C per year for the moderate-emission scenario and +0.035°C per year for high-emission trajectories. Rainfall projections followed empirically derived trends, showing increasing variability, changes in seasonal distribution and heightened likelihood of intense episodic rainfall events. These climatic insights were interpreted in relation to Kilifi's hazard profile, reinforcing the need for enhanced water security, climate-resilient agricultural practices, improved coastal and marine ecosystem management, stronger infrastructure planning and expanded early warning systems.

1.2.6 Data Analysis and Reporting

Quantitative data from the household surveys were cleaned, coded and analyzed using Microsoft Excel and Python. Analysis focused on frequencies, proportions, cross-tabulations and simple correlations across gender, age group, livelihood type and vulnerability status. These results provided empirical confirmation of patterns emerging from qualitative insights.

Qualitative data from WAPs, FGDs, PCVA exercises and transect walks were transcribed and thematically coded to organise evidence under key domains such as hazard exposure, livelihood

impacts, gender dynamics, coping strategies, institutional barriers and opportunities. Outputs from PCVA tools; maps, timelines, calendars, clocks, matrices and impact chains, were digitized and integrated with NDVI layers and climate trend visualizations to produce synthesized analytical outputs. Triangulation across all sources ensured internal consistency and improved the robustness of the final interpretation. Draft findings were compiled for review by Plan International Kenya in line with expected deliverables.

1.2.7 Validation

The draft findings were presented in county-level validation workshop held in Kilifi on 20th November 2025. This workshop provided a platform for county technical departments, CCCPC/WCCPC members, civil society, COSME partners and community representatives to review the PCVA results and refine the draft adaptation strategies. At ward level, stakeholders from the three wards reviewed hazard rankings, adaptation priorities and institutional roles, and provided further refinement on feasibility, gender inclusion, sequencing and resource implications. Inputs from these processes were incorporated into the final Ward Adaptation Plan to ensure scientific accuracy, community legitimacy and institutional alignment.

1.3 Linkages to other planning processes

The Kilifi County Climate Change Adaptation Action Plan (CCAAP 2025–2030) builds on existing national, county, and ward-level planning frameworks to enhance coherence, coordination, and integration of climate adaptation across governance levels. The plan draws directly from the Participatory Climate Vulnerability and Risk Assessments (PCVAs) conducted in fourteen villages across Mnarani, Mwarakaya and Sokoke Wards and from the resulting Ward Adaptation Plans (WAPs) developed under the COSME Project, which provided detailed understanding of climate risks, vulnerabilities, and adaptive capacities within coastal communities. At the national level, the plan aligns with the National Adaptation Plan (NAP 2015–2030), which operationalizes the National Climate Change Action Plan (NCCAP) by embedding adaptation across all sectors of development through a multi-sectoral and multi-level governance framework that recognizes climate change as a cross-cutting issue influencing Kenya's economic growth, social well-being, and environmental stability. At the county level, the CCAAP complements the Kilifi County Climate Change Action Plan (CCAP 2023–2032), which localizes the NAP's priorities within Kilifi's ecological and socio-economic context and integrates adaptation objectives into the County Integrated Development Plan (CIDP) and Annual Development Plans (ADPs). CCAAP translates these strategic directions into locally driven, gender-responsive adaptation measures that address community priorities identified through participatory assessments. By linking ward-level findings to county and national frameworks, the plan promotes vertical integration of climate action, ensuring that localized initiatives contribute to broader resilience and sustainable development outcomes across Kilifi County.

2 Chapter 2: Coordination of Adaptation Actions

The Climate Change Adaptation Action Plan is guided by a multi-level governance framework spanning national, county, ward, and community institutions. At the national level, Kenya's policy and legal framework establishes the mandates and coordination structures for climate action. The Climate Change Act (2023) and the National Adaptation Plan (NAP) 2015–2030 set the strategic direction for adaptation, emphasizing mainstreaming of climate resilience in all sectors and levels. The National Climate Change Action Plan (NCCAP) 2023–2027 operationalizes these priorities into programs and financing mechanisms, such as the Financing Locally Led Climate Action (FLLoCA) program which channels performance-based climate finance to counties and communities. A National Climate Change Council and Climate Change Directorate provide high-level oversight and technical guidance, while specialized national agencies support implementation: for example, the Kenya Meteorological Department (KMD) provides localized forecasts and early warnings; the National Drought Management Authority (NDMA) leads drought early warning and response; the Kenya Forest Service (KFS) and Kenya Wildlife Service (KWS) support forest conservation, ecosystem restoration, and human–wildlife conflict management; and the National Environment Management Authority (NEMA) ensures environmental safeguards in climate projects. These bodies work in concert with county governments to ensure that national adaptation objectives (e.g. integrated water resource management, climate-smart agriculture, disaster risk reduction, protection of forests and coastal zones) are translated into locally relevant actions.

At the county level, the Kilifi County Government serves as the critical bridge between national policies and community needs. The County Government has institutionalized climate change within its structures in line with the Climate Change Act. Notably, the Kilifi County Climate Change Act (2021) established a dedicated County Climate Change Unit and a County Climate Change Fund (CCCF) to coordinate climate programs and finance adaptation projects. Climate change is now mainstreamed in the County Integrated Development Plan (CIDP) 2023–2027 and sectoral plans, meaning departments such as Water, Agriculture, Livestock and Fisheries, Roads and Public Works, Health, and Environment must integrate resilience and low-carbon measures into their annual budgets and activities. A County Climate Change Action Plan (2023–2027) further details county-specific targets (e.g. on water security, resilient agriculture, ecosystem restoration, infrastructure protection) and provides an enabling framework for ward-level proposals to access climate finance. County coordination platforms, including the County Steering Group (for development and humanitarian partners coordination) and the County Disaster Management Committee, are increasingly utilized to plan risk reduction and anticipatory actions based on seasonal climate forecasts. Through these mechanisms, the county government aligns external donor/NGO interventions with public investments, avoids duplication, and ensures climate actions are coherent with both national policy and community priorities.

At the ward and community level, implementation and local planning are anchored by Ward Climate Change Planning Committees (WCCPCs) and related community institutions. Each ward (such as Mnarani, Mwarakaya and Sokoke) has established a WCCPC as an inclusive platform for identifying needs, prioritizing adaptation projects, and monitoring progress. The WCCPC is chaired by the Ward Administrator and comprises representatives of local government (e.g. ward Agricultural officers, Public Health officers), community groups (Beach Management Units for fisheries, Water Resource User Associations, Community-Based Organizations), women and youth representatives, persons with disabilities, and respected community leaders. This broad composition mirrors the NAP's emphasis on locally led adaptation and ensures that planning is participatory and considers gender and social inclusion. The WCCPC works closely with issue-specific committees (water user committees manage boreholes and water points; disaster volunteer teams coordinate local emergency response;

forest/pasture management groups oversee ecosystem activities). Historically, many of these community efforts operated in silos or informally; the WCCPC now provides a structured mechanism for coordination at the ward level, facilitating information sharing and joint action across villages and sectors.

To link ward efforts with county support, clear institutional interfaces have been established. A County Climate Change Officer is designated to liaise with each WCCPC, attending ward meetings and ensuring ward priorities are conveyed to the County Climate Change Unit for integration into county programs and CCCF funding pipelines. The Ward Administrator and WCCPC produce quarterly progress reports on adaptation projects, which feed upward to the County Climate Change Unit, capturing outputs, challenges, gender-disaggregated impacts, and resource needs. This reporting routine embeds accountability and allows the county to monitor ward-level progress in line with county and national indicators (e.g. those of FLLoCA and the Climate Change Act). Multi-stakeholder forums are also convened: each ward will hold a biannual Ward Climate Stakeholder Forum under the WCCPC, bringing together county departmental officers, national agencies operating locally, NGOs, CBOs, private sector, and research institutions active in that ward. This forum maps who is doing what and where, aligns technical standards, and resolves any implementation bottlenecks collaboratively. These arrangements ensure vertical integration (ward plans informing county budgets and vice versa) and horizontal coordination (all local actors working in synergy).

2.1 Policy alignment and implementation roles

The CCAAP is anchored in existing policy frameworks. It operationalizes Kenya's NAP 2015–2030 priorities by focusing on the very sectors highlighted at national level, water, agriculture, health, infrastructure, ecosystems, and disaster risk management, but tailors actions to Kilifi's local context. It also aligns with the NCCAP 2023–2027, tapping into national programs like FLLoCA and the National Drought Emergency Fund to support local actions. The plan is consistent with the Kilifi County Climate Change Action Plan (2023–2027) and the Climate Change Act (2021), leveraging the County Climate Change Fund mechanism to fund community-prioritized projects. Climate change is integrated in the CIDP 2023–2027, meaning this adaptation plan will be mainstreamed into the county's development agenda and annual development plans. Finally, implementation will be guided by institutional roles as defined in law: the WCCPC leading community mobilization and prioritization, county departments providing technical support and funding through sector budgets, and national agencies and development partners providing specialized inputs (early warning information, climate finance, etc.). Table 1 below summarizes the key institutions and their roles in implementing the adaptation actions. The CCAAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks.

Table 1: Institutional Roles in Implementation of CCAAP

Institution / Actor	Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead community engagement, planning and prioritization of ward-level adaptation actions; ensure projects are inclusive (gender and youth); coordinate implementation at ward level.
Kilifi County Government Departments	Integrate ward adaptation priorities into sector plans and budgets (Water, Agriculture, Livestock, Fisheries, Roads, Health, Environment); provide technical expertise and co-funding; oversee infrastructure works and service delivery.

Institution / Actor	Roles and Responsibilities
County Climate Change Unit (CCU)	Coordinate CCAAP implementation county-wide; manage the County Climate Change Fund (CCCF) allocations; consolidate M&E reporting from wards; liaise with National Climate Change Directorate on finance and reporting compliance.
National Drought Management Authority (NDMA)	Monitor drought conditions; issue early warnings; trigger drought emergency funds for interventions (water trucking, food relief) when thresholds are reached; support community drought contingency planning.
Kenya Meteorological Department (KMD)	Provide localized weather forecasts and seasonal climate outlooks; issue early warnings for extreme weather (heavy rain, storms); support establishment of ward-level weather monitoring (rain gauges) and advisories for farmers and fishers.
Kenya Forest Service (KFS)	Advise on forest restoration and agroforestry; support tree nursery establishment and management (including mangroves); enforce protection of gazetted forests; assist communities in implementing reforestation and woodlot projects.
Kenya Wildlife Service (KWS)	Handle human–wildlife conflict mitigation (e.g. respond to wildlife incursions, problem animal control); support community education on wildlife safety; facilitate compensation claims for wildlife damage; help establish wildlife deterrent measures near parks/reserves.
Beach Management Units (BMUs) & Fisheries Service	Co-manage coastal and marine resources; implement sustainable fishing practices; participate in mangrove conservation and reef protection to buffer against sea-level rise and storms; support alternative livelihoods for fisherfolk (e.g. mariculture).
Community-Based Organizations (CBOs) & Self-Help Groups	Implement community-led projects (water harvesting, tree planting, soil conservation, savings groups); mobilize local resources and labor; ensure inclusion of women, youth, and marginalized in project benefits.
Development Partners & NGOs (e.g. World Bank FLLoCA, UNDP, WWF, Plan International)	Provide funding grants and technical assistance for local adaptation projects; build capacity (training, tools) for communities and county staff; support innovation (e.g. climate-smart agriculture pilots, ecosystem-based adaptation) and document lessons for scaling up.
Private Sector & Cooperatives	Invest in climate-smart value chains (e.g. drought-tolerant seeds, solar irrigation kits); offer products like weather-index insurance for farmers; participate in Public-Private Partnerships for resilient infrastructure (water supply, renewable energy); support community projects through corporate social responsibility.
Academic & Research Institutions (e.g. KMFRI, Universities)	Generate localized climate data and risk analyses; monitor ecosystem changes (marine and terrestrial); pilot new adaptation technologies (e.g. salt-tolerant crops, erosion control methods); evaluate adaptation outcomes and advise on improvements; support climate knowledge management and learning.

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

Kilifi County experiences multiple and interlinked climatic hazards that have intensified over the past decades, directly impacting livelihoods, ecosystems, and infrastructure. The Participatory Climate Vulnerability and Risk Assessments (PCVAs) conducted across fourteen villages namely Takaungu, Mkwajuni-Takaungu, Vuma, Vijiweni, Midzimitsano and Timboni in Mnarani Ward: Mbuyuni, Ondoni, Vwevwesi, Silala in Mwarakaya Ward and Dikunze, Ndungumnani, Mwele, Midugani in Sokoke Ward, revealed that the county faces recurrent droughts, floods, pests and diseases, environmental degradation, strong winds, and human–wildlife conflicts (HWCs). These hazards are influenced by shifting climate patterns characterized by rising mean annual temperatures (increasing by approximately 0.3°C per decade), irregular and declining rainfall reliability, and prolonged dry spells particularly between July and October. Flood events have become more frequent during the long rains (March–May), leading to displacement and loss of crops and livestock, while extended droughts have intensified water scarcity and food insecurity. Along the coastal belt, sea level rise and coastal erosion threaten settlements and productive areas, with low-lying regions such as Takaungu increasingly experiencing saltwater intrusion and shoreline retreat.

The cumulative effect of these hazards has eroded livelihood security and degraded key ecosystems such as forests, mangroves and kayas thereby increasing competition over scarce resources. The PCVA findings underscore a county-wide trend of increased climate variability where increased frequency of hazards and intensity continue to amplify vulnerabilities across social groups, particularly women, youth and Persons Abled Differently (PAD)necessitating integrated, gender responsive and ecosystem-based adaptation interventions.

3.2 Current climate trends

Kilifi County experiences a tropical climate characterized by bimodal rainfall patterns and relatively high temperatures throughout the year. The analysis of historical and near-present climatic parameters, including rainfall, potential evapotranspiration (PET), and actual evapotranspiration (AET), indicates that rainfall amounts have shown a modest upward trend, while PET and AET values reflect strong evaporative demand typical of semi-humid coastal systems. Under the moderate emission scenario (SSP2–4.5), rainfall is projected to increase slightly from 1,069.0 mm to 1,078.8 mm, while PET shows a marginal decline from 1,686.6 mm to 1,683.9 mm. Actual evapotranspiration (AET) also shows a corresponding increase from 1,932.7 mm to 1,936.2 mm over the same period (Figure 4). This subtle rise in rainfall marginally offsets the evaporative losses, suggesting a moderate improvement in soil moisture availability and vegetation water balance. The observed relationship between rainfall and evapotranspiration demonstrates the interdependence between moisture supply and atmospheric water demand, where even small increases in rainfall can influence the hydrological equilibrium and agricultural productivity.

Temperature trends further reinforce the warming tendencies across the county. Under SSP2–4.5, mean temperatures are projected to rise from 26.68 °C to 27.04 °C, while land surface temperature (LST) increases from 36.15 °C to 36.37 °C (Figure 5). This consistent thermal rise points to a gradual intensification of surface heating, which may elevate evapotranspiration rates and consequently increase crop-water demand. The warming trajectory implies growing heat stress on vegetation, livestock, and human systems, with implications for water resource management and agricultural scheduling.

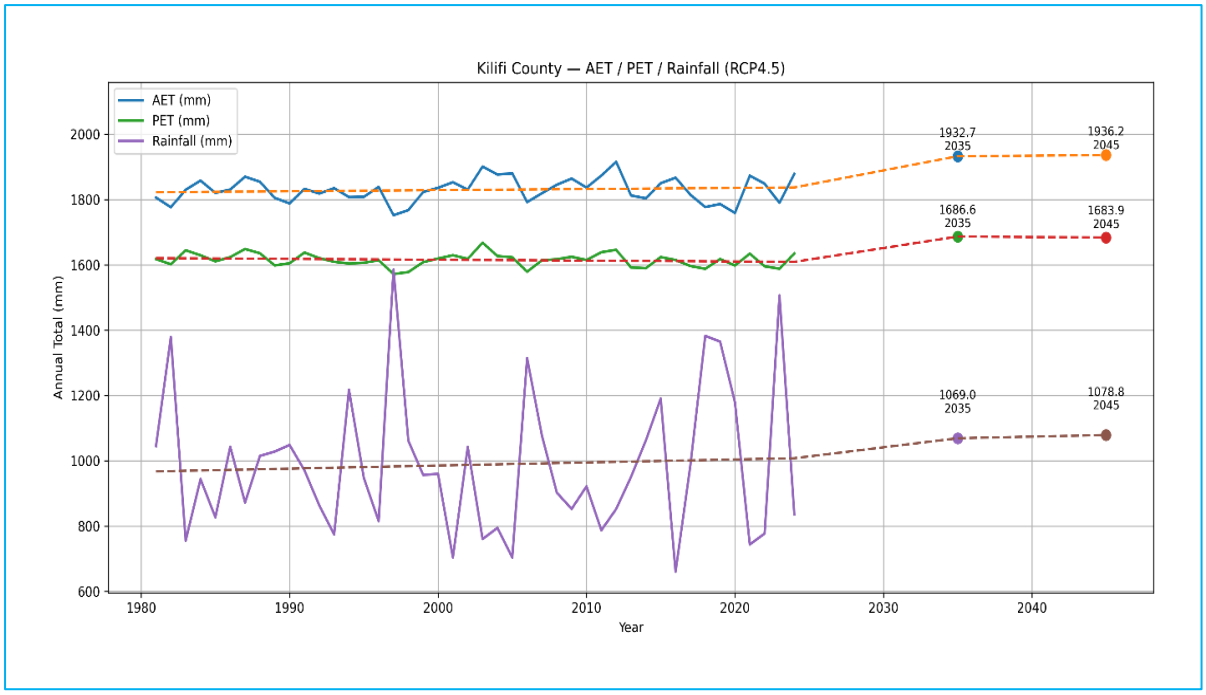

Figure 4: SSP2 4.5 Rainfall, AET and PET Projections (Source: Chirps database and ERA5-Ag, 2022)

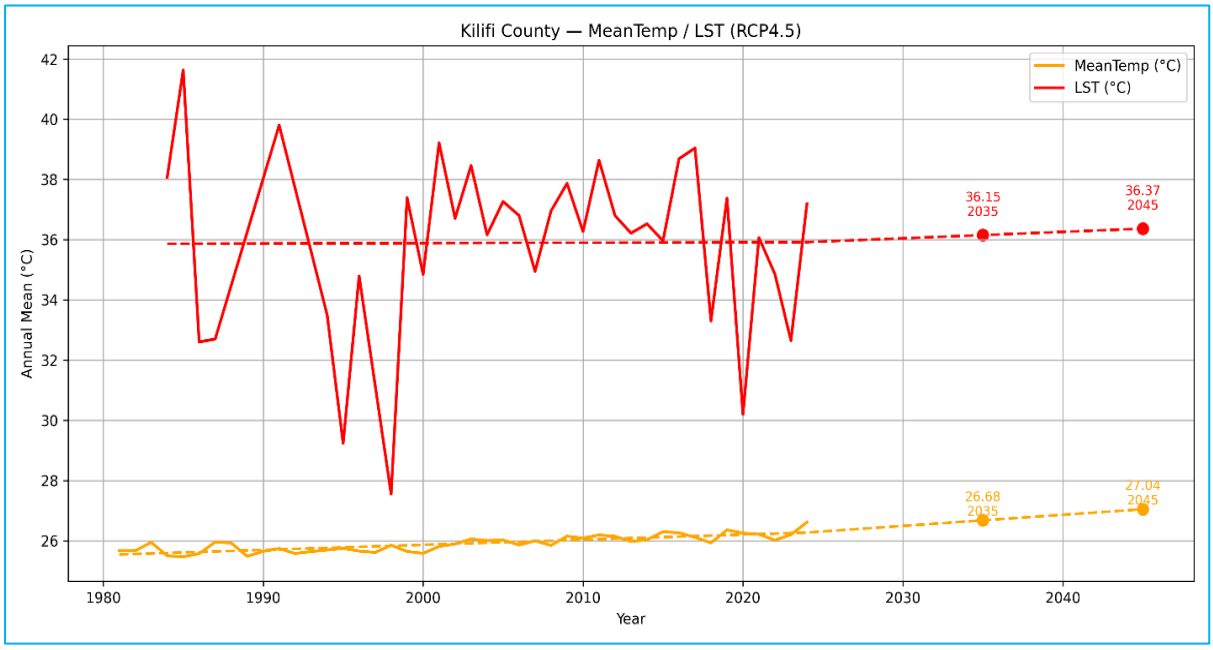

Figure 5: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.3 Future climatic trends

Projected climate scenarios for Kilifi County, developed from CHIRPS rainfall and ERA5-Ag evapotranspiration and temperature models, suggest an amplification of current trends under both moderate (SSP2–4.5) and high-emission (SSP5–8.5) pathways. Under the high-emission scenario (SSP5–8.5), which assumes accelerated greenhouse gas concentrations, the county is projected to experience more pronounced climatic changes (Figure 6). Rainfall is expected to rise from 1,119.9 mm in 2035 to approximately 1,130.2 mm by 2045, reflecting a small but consistent increase. However, despite this growth, the PET remains elevated, though slightly declining from 1,767.0 mm to 1,764.0 mm, indicating sustained atmospheric water demand.

Meanwhile, AET is projected to increase from 2,024.7 mm to 2,028.4 mm, illustrating enhanced hydrological fluxes as a result of intensified rainfall–temperature interactions. These shifts suggest that while precipitation may improve water availability, the overall balance may still favor high evapotranspiration losses, particularly during extended dry spells, leading to periodic moisture stress and spatially variable water deficits.

Similarly, temperature projections under the SSP5–8.5 pathway indicate accelerated warming, with mean temperature expected to rise from 26.84 °C to 27.36 °C, and land surface temperature from 36.32 °C to 36.68 °C (Figure 7). Such a pattern represents heightened heat flux and increased surface radiation, which may exacerbate land degradation, reduce soil moisture retention, and intensify evapotranspiration potential. The combined effect of rising temperatures and fluctuating rainfall patterns points to an increasingly volatile climate system, where extreme events such as prolonged droughts, heatwaves, and erratic rainfall may become more frequent.

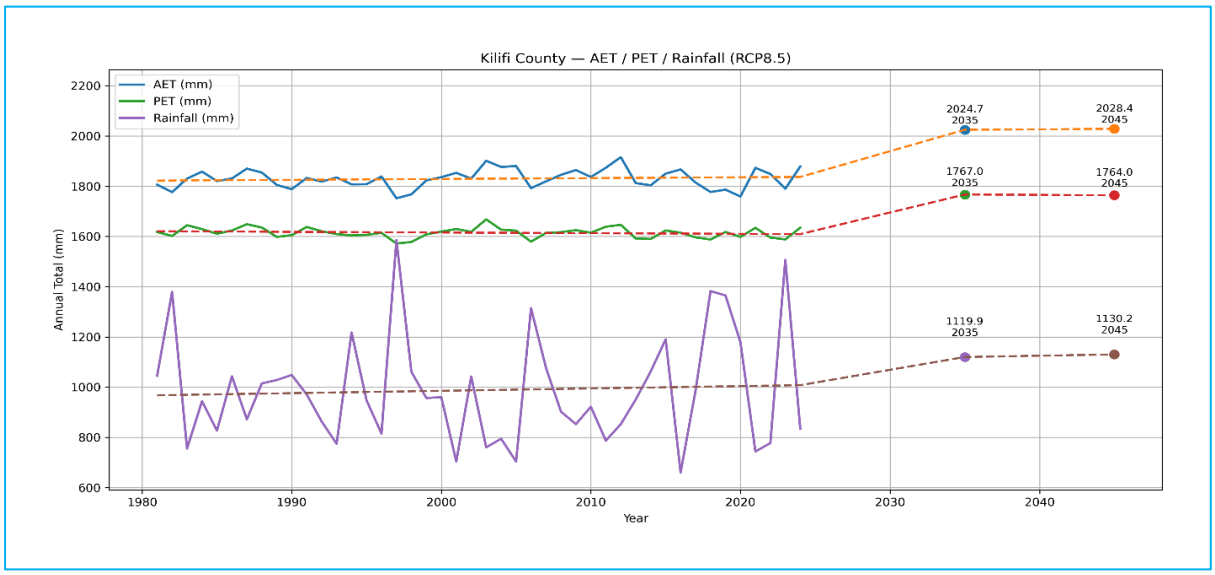

Figure 6: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

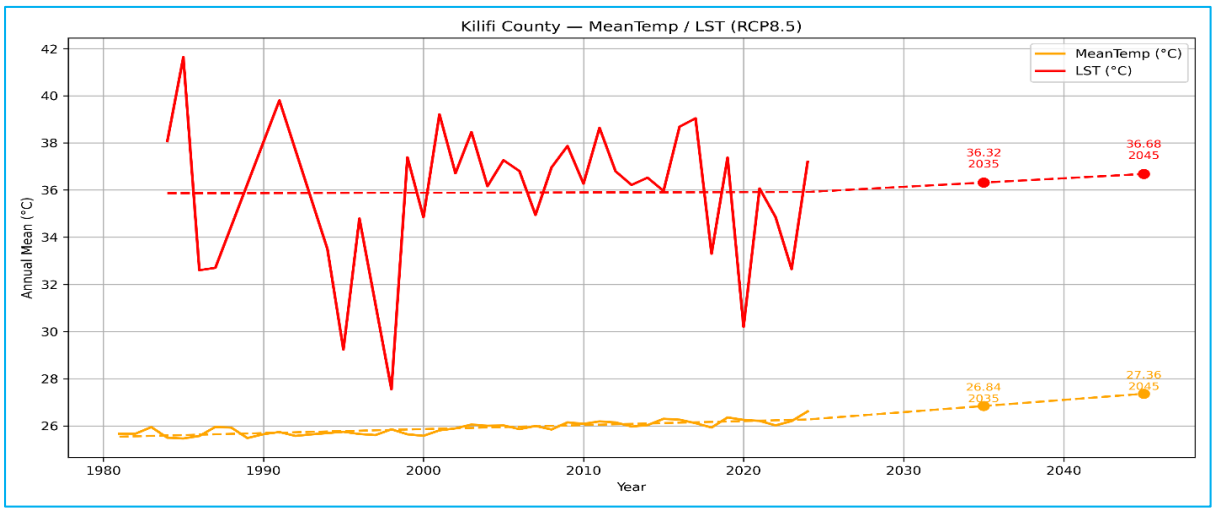

Figure 7: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.4 Climate hazards and vulnerability

3.4.1 Drought

Drought is the most recurrent and economically devastating climatic hazard in Kilifi County, exerting sustained pressure on ecosystems and livelihoods. Historical records from 1930 to 2025 reveal a steady rise in drought frequency and intensity, with major events in 1930, 1947, 1948, 1960, 1978–1982, 1997–1998, 2012–2015, 2017, and 2024. Increasingly prolonged dry spells, delayed rainfall onset, and early cessation have reduced growing periods, caused recurrent crop failures, and heightened stress on terrestrial and coastal ecosystems. Prolonged drought has driven severe vegetation loss and biodiversity decline in areas such as Kaya Chonyi, Arabuko Sokoke Forest, Kambi ya Shetani, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni. The drying of riverine systems like Taratibu, Kombeni, and Mavueni, along with water pans such as Mbogolo, Mahara, and Mabirikani, has further reduced soil fertility, agricultural productivity, and household incomes. Mangrove ecosystems along Takaungu Creek are also affected, with regeneration failure disrupting fisheries. These stresses have pushed communities toward unsustainable coping strategies, sand harvesting, charcoal burning, and overexploitation of forest resources, deepening both environmental degradation and socio-economic vulnerability.

Ecologically, recurrent droughts have reshaped the structure and function of key ecosystems, as shown by variations in Normalized Difference Vegetation Index (NDVI) values. In Kaya Chonyi (Mwarakaya Ward), dry-season NDVI values (0.30–0.49) indicate moderate vegetation stress, while wet-season values (0.24–0.60) reflect recovery with increased rainfall. Similarly, Mwangea Hills (Sokoke Ward) show NDVI ranges of 0.14–0.55 during dry periods and 0.03–0.58 in wet seasons, indicating vegetation revival with improved moisture. At Takaungu Creek (Mnarani Ward), mangroves display high seasonal sensitivity, with NDVI values of 0.24–0.49 in dry periods and 0.06–0.51 in wet seasons, where flooding and sedimentation hinder regeneration. Collectively, these patterns reveal that rainfall variability and prolonged droughts are key drivers of vegetation stress, declining greenness, and ecosystem degradation across Kilifi County.

3.4.1.1 Exposure to Drought

Exposure to drought varies across Kilifi County, shaped by livelihood dependence, ecosystem conditions, and water access. Drought exposure in the county is widespread, affecting both ecological and livelihood systems. In Mwarakaya Ward, hotspots such as Chengoni, Chije, Mbuyuni, Kizingo, Mitulani, Bandarasalama, and Mwarakaya experience frequent drying of water pans, boreholes, and shallow wells due to declining groundwater levels. Similarly, in Sokoke Ward, key ecosystems, including Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni, face prolonged dry spells and rising temperatures. Reduced canopy cover and slowed regeneration in Arabuko Sokoke Forest have increased fire risks and threatened endemic species, while Mwangea Hills suffers erosion and drying of seasonal streams, reducing downstream water supply. In the entire county, declining water availability heightens competition, social tensions, and reduced farm productivity. The Kaya forests (Chonyi, Godoma, and Kadzitsoni) also experience soil moisture loss, vegetation dieback, and biodiversity decline, disrupting cultural heritage. The mangrove ecosystems located along Takaungu Creek are highly exposed during prolonged dry spells limiting mangrove regeneration affecting fishing activities that sustain some local households. County wide, pasture and water scarcity have led to livestock mortality and reduced milk yields, compounding livelihood losses. These dynamics underscore the interconnectedness of drought impacts across Kilifi County, driving poverty, food insecurity, and environmental degradation.

3.4.1.2 Socio-Demographic Vulnerability to Drought

Socio-demographic vulnerability to drought is pronounced among women, youth, elderly persons, and PAD whose livelihoods depend on climate-sensitive resources. Women bear the

brunt of domestic water scarcity, fuelwood collection, and food production. As household food insecurity rises, they face increased unpaid labour and reduced participation in income-generating activities. Youth engaged in casual farm work and fishing experience loss of income as agricultural productivity declines. Elderly persons and PAD face mobility challenges that limit access to distant water sources and relief services. Economic vulnerability compounds the challenge, most households earn less than KES 13,000 monthly, limiting investment in irrigation or water storage. Weak access to extension services and early warning information further limits preparedness. Across the wards, women's limited land ownership and access to credit hinder adoption of drought-tolerant crops or livestock breeds, perpetuating gendered vulnerability. (Source: PCVA, 2025)

3.4.2 Floods

Historical flood records between 1930 and 2025 indicate major flood occurrences in 1930, 1961, 1965, 1972, 1997, 2018 and 2023, translating to an average recurrence interval of about one significant flood every six to thirteen years. The frequency and magnitude of floods have intensified in recent decades as a result of increased rainfall variability, catchment degradation, and expansion of settlements into flood-prone zones. Mwele, Ondoni, Silala and Mbuyuni villages are the most exposed, experiencing recurring damage to farmlands, settlements and infrastructure. Floods have also affected key ecosystems such as Arabuko Sokoke Forest, Kaya Godoma and Kaya Kadzitsoni, leading to soil erosion, excessive surface runoff, and loss of vegetation cover, which in turn undermine their ecological roles in soil stabilization, groundwater recharge, and hydrological regulation.

The effects of floods on natural and livelihood resources are widespread and multifaceted. Kaya Chonyi experience soil erosion, excessive surface runoff, and loss of ground cover, undermining ecological functions such as soil stabilization, groundwater recharge, and hydrological regulation. Along the riverine ecosystems of River Mwanzai and Mto Mkuu flooding results in severe land degradation, destruction of adjacent farmlands and settlements as well as the loss of livelihoods. Mkwajuni, Midzi Mitsano, Vijiweni, Vuma and Timboni which are predominantly situated along natural drainage channels, floodplains, and degraded catchment zones are also most affected areas.

3.4.2.1 Exposure to Floods

Flood exposure across the wards is widespread, unevenly distributed and closely linked to the riverine ecosystems of River Mwanzia, Mto Mkuu, Milore, Rare, Muriachakwe, Katangani, and Nyare. In Mwarakaya, the most flood-prone areas, Ondoni, Mbuyuni, and Silala, experience extensive inundation that destroys farmlands, reduces crop yields, and erodes access roads, disrupting mobility and isolating communities during peak rainfall seasons. Similarly, in Sokoke Ward, recurrent flooding in low-lying and riparian zones leads to loss of fertile topsoil, destruction of croplands, and property damage, particularly in settlements lacking adequate stormwater infrastructure. These impacts collectively compromise food production, increase displacement, and intensify food insecurity among households reliant on seasonal farming. Coastal and riparian zones along Takaungu–Kivukoni, Mbogolo, and Vuma host vital mangrove ecosystems that support fisheries, protect shorelines, and store carbon. However, recurrent flooding and catchment degradation have led to increased sedimentation, saltwater intrusion, and mangrove die-back, reducing regeneration and altering species composition. These impacts have also extended inland, affecting terrestrial forests in upland areas such as Midzi Mitsano and Mkwajuni through cumulative ecosystem degradation and changing hydrological patterns.

3.4.2.2 Socio-Demographic Vulnerability to Floods

Flood vulnerability in Kilifi is shaped by socio-economic status, gender, and settlement patterns. Women, particularly those engaged in informal trade and caregiving, experience significant losses during floods due to destruction of property and disruption of household water and sanitation facilities. Children face health risks from contaminated water and unsafe play environments, while elderly persons and PAD face challenges evacuating flooded areas. Poor households, especially those living near riverbanks in makeshift structures, lack financial capacity to rebuild, trapping them in poverty cycles. Female-headed households bear additional responsibilities for food, health, and childcare during recovery. Youth working in fishing and seaweed farming suffer income interruptions as production halts. (Source: PCVA, 2025)

3.4.3 Pests and Diseases

Pest and disease outbreaks have become increasingly frequent climate-induced hazards across wards, posing major threats to ecosystems, livelihoods and food security. Historical records from 1978 to 2025 show that outbreaks occur at least once every two years, with notable peaks between 1978–1998, 2011, 2013, 2017, 2019, 2020, 2022 and 2025. Their rising frequency and spread are closely linked to erratic rainfall, higher temperatures, and prolonged dry spells that weaken crop and livestock resilience. Across the county, wards have experienced widespread crop infestations, including armyworms and stalk borer on maize, fruit flies on mangoes, cassava mosaic virus, aphids, and whiteflies, resulting in significant yield losses and reduced household incomes. Livestock diseases such as foot-and-mouth and tick-borne infections (east coast fever) have further exacerbated economic impacts. Natural ecosystems have also been severely affected; key forests such as Kaya Chonyi, Arabuko Sokoke, Godoma, and Kadzitsoni have suffered pest-related leaf defoliation and fungal infections, leading to reduced photosynthetic capacity, impaired regeneration and biodiversity loss. These disruptions undermine vital ecosystem services, such as soil stabilization, water regulation and microclimate moderation, further diminishing the ecological resilience of wards.

3.4.3.1 Exposure to Pests and Diseases

Exposure to pests and diseases across the wards is widespread, affecting all villages, ecosystems, and livelihood systems. Wards experience recurrent outbreaks that impact crops, livestock, and human health, driven by changing climatic conditions, particularly fluctuations in temperature, rainfall and humidity. In Mwarakaya, periodic pest invasions such as fall armyworms and fruit flies have devastated farmlands in Vwevwesi, Ondoni, and Mbuyuni, while livestock diseases including foot-and-mouth and tick-borne infections have increased following alternating drought and wet seasons. Similarly, in Sokoke, pest infestations such as locusts and armyworms have caused severe damage to farmlands in Midugani and Mwele, with comparable increases in livestock disease outbreaks. Rising temperatures, erratic rainfall, and recurrent flooding have created favourable breeding conditions for vector- and waterborne diseases such as malaria and cholera. Changes in moisture and temperature regimes have also had significant impacts on mangrove and terrestrial forest ecosystems, accelerating defoliation and reducing natural regeneration capacity in Mnarani Ward. Infestations such as wood-boring insects degrade mangrove stands and terrestrial vegetation that serve as natural protective buffers against flooding and erosion.

3.4.3.2 Socio-Demographic Vulnerability to Pests and Diseases

Smallholder farmers and pastoralists are the most affected, with women particularly vulnerable as they manage subsistence farms and small livestock susceptible to outbreaks such as locusts, fall armyworms, and fruit flies. These infestations reduce harvests, undermining income and food security, while limited access to pest control technologies, extension services, and credit restricts adaptation. Youth relying on casual agricultural labour face income loss and migration pressures during outbreaks, while elderly persons and persons abled differently (PAD) are

constrained by physical limitations. In villages such as Midugani, Mwele, Mbuyuni, Silala, and Vwevwesi, livestock diseases including foot-and-mouth and tick-borne infections erode household assets, especially among poorer families lacking veterinary support. Children and the elderly are also prone to secondary illnesses like malaria and dengue fever linked to pest proliferation.

3.4.4 Human–Wildlife Conflicts

Human–Wildlife Conflicts (HWC) have become an increasingly significant climate-related hazard in Kilifi County, particularly in areas bordering degraded forest ecosystems and wildlife corridors. In Mwarakaya, incidents, historically rare, now occur roughly every five to six years, with a notable surge in 2025 when several cases were reported in Ondoni and Vwevwesi villages adjacent to Kaya Chonyi. In Sokoke Ward, conflicts are more frequent, occurring almost annually, especially in areas forming ecological linkages between Arabuko-Sokoke Forest and Tsavo East National Park, which serve as key dispersal and migration routes for wildlife. The rise in these conflicts is closely tied to deforestation, agricultural expansion, and changing rainfall patterns that reduce water and forage availability within forest habitats, pushing wildlife such as baboons, monkeys, and wild pigs into nearby farmlands and settlements.

The degradation of Arabuko-Sokoke Forest, Kaya Godoma, Kaya Kadzitsoni, and Kaya Chonyi has increased human–wildlife conflicts (HWC). Loss of forest cover, declining water sources, and reduced forage have disrupted wildlife habitats, driving animals into nearby farms and settlements in search of food and water. Elephants destroy maize and cassava, while baboons and monkeys raid farms, causing frequent yield losses; livestock such as goats and poultry are also preyed upon, worsening economic losses.

3.4.4.1 Exposure to Human–Wildlife Conflicts

Exposure to human–wildlife conflicts (HWC) is highest in communities near degraded forest habitats and wildlife corridors, where declining vegetation cover, shrinking water sources, and farmland expansion have intensified human–wildlife interactions. Villages such as Vwevwesi and Ondoni, located adjacent to Kaya Chonyi, experience frequent crop raids and farm destruction as wild pigs, baboons, and monkeys forage beyond forest boundaries, particularly during dry seasons. Exposure is also greatest around Dkunze Village bordering the Arabuko-Sokoke Forest, where degradation of Arabuko-Sokoke, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni has weakened natural barriers and food availability, driving elephants, baboons, and monkeys into croplands.

3.4.4.2 Socio-Demographic Vulnerability to HWCs

Smallholder farmers and pastoralists face frequent crop raids by elephants, baboons, monkeys, and wild pigs, as well as livestock predation, which erode food security and income stability. Women are most exposed due to their daily roles in farming, water collection, and firewood gathering near forest edges, while children, widows, the elderly, and PADs face heightened risks from limited mobility and safety measures. The destruction of crops and livestock forces households to buy food, rely on aid, or migrate temporarily, disrupting livelihoods and deepening poverty.

3.4.5 Strong Winds

Strong winds have become a recurrent climate hazard, causing extensive damage to property, vegetation, and livelihoods. Between 2023 and 2025, these events occurred almost annually, with a major outbreak in 2024 that severely affected areas such as Ndugumnani, Mwele, Dzikudze, Ndigiria, Mapotea, Mitedzini, and Mrima wa Ndege. Sparse vegetation, open landscapes, and poorly constructed houses have heightened exposure to wind damage and soil erosion. The strong gusts uproot trees in areas like Arabuko Sokoke Forest, Mwangea Hills,

Kaya Godoma, and Kaya Kadzitsoni, disrupting ecosystems, reducing biodiversity, and accelerating land degradation. Crops such as maize, cassava, and coconuts are often flattened, while granaries, livestock shelters, and beehives are destroyed, leading to yield losses and economic hardship. In Vuma, extensive deforestation and mining by Mombasa Cement have worsened the impacts by leaving land bare and prone to wind erosion. The 2023 whirlwind caused severe destruction to schools, plantations, and infrastructure.

3.4.5.1 Exposure to Strong Winds

Exposure to strong winds is widespread, with the most severe impacts reported in open and low-lying areas such as Mwele, Ndugumnani, Dzikunze, Midzedzini, Mrima wa Ndege, and Vuma, where sparse vegetation and weak housing structures heighten vulnerability. Strong winds frequently blow off roofs, uproot trees, and destroy farmlands, especially in exposed plains where crops, young seedlings, and fruit trees are lost, undermining both food security and household income. The situation is worsened by widespread deforestation and agricultural land clearing, which have eliminated natural windbreaks, while poorly maintained community buildings, schools, and markets suffer recurrent damage that disrupts education and livelihoods. In Vuma, open mining zones and upland ridges stripped of vegetation have intensified wind hazards, eroding soils and damaging plantations, particularly coconut and cashew crops. Most households, built from lightweight materials such as makuti, iron sheets, and mud, are highly susceptible to damage, with limited access to resources or technical capacity for constructing wind-resistant structures.

3.4.5.2 Socio-Demographic Vulnerability to Strong Winds

Women face heightened risks as they work near weak homesteads, farms and markets, while the destruction of food stores and assets deepens economic strain. Youth in casual work and vending lose income when markets and roads are damaged, whereas the elderly and PADs struggle to evacuate or rebuild after events. In Vuma, women and youth in coconut processing and informal trade suffer heavy losses, and men in mining and fishing face physical danger from debris and turbulent conditions. Recurrent wind damage has weakened household resilience, demanding integrated interventions such as vegetation restoration, mine rehabilitation, and promotion of wind-resistant construction to enhance adaptive capacity.

3.4.6 Environmental Degradation

Environmental degradation in the wards has become an escalating hazard driven by unsustainable human activities such as tree felling for charcoal, timber, and fuelwood, alongside population growth, agricultural expansion, and urbanization. Historical assessments between 1987 and 2025 indicate that major degradation episodes occurred roughly every five years, with notable peaks in 1997, 2005–2010, and 2024, coinciding with periods of rapid settlement expansion and intensified resource exploitation, particularly around Dzikudze, Takaungu, and Vuma. Over time, localized disturbances have evolved into widespread ecosystem decline affecting forests, riverbanks, wetlands, and coastal zones. The increasing demand for fuelwood, fisheries, and tourism-related products has accelerated deforestation within mangrove and terrestrial ecosystems, while inadequate waste management and unregulated dumping have worsened pollution and debris accumulation in aquatic environments.

The Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni have faced severe degradation from tree felling for fuelwood, charcoal, and construction materials, disrupting hydrological cycles, reducing groundwater recharge, and intensifying drought and flood risks. The loss of forest and mangrove cover has led to declining biodiversity, siltation of water bodies, reduced fish stocks and accelerated coastal erosion.

3.4.6.1 Exposure to Environmental Degradation

Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma and Kaya Kadzitsoni have undergone severe degradation from unregulated farming, fuelwood extraction and settlement expansion, resulting in soil erosion, declining fertility, and disrupted hydrological cycles. The degradation of Arabuko Sokoke Forest has accelerated biodiversity loss and heightened vulnerability to floods and droughts due to reduced groundwater recharge. Likewise, mangrove loss in Takaungu, Vuma, and Timboni has exposed coastal communities to tidal surges, erosion, and dwindling fisheries. Uncontrolled mangrove cutting, pollution, and encroachment have weakened the natural resilience of these ecosystems.

3.4.6.2 Socio-Demographic Vulnerability to Environmental Degradation

Socio-demographic vulnerability stems from gender roles, livelihood dependence, income inequality and unequal resource access. Women, smallholder farmers, and youth are most affected, relying heavily on forests for wood fuel, and income. Deforestation in Arabuko Sokoke Forest forces women to walk long distances for firewood and water, increasing physical strain and time poverty. Farmers face declining soil fertility and disrupted water cycles, reducing crop yields and food security. Youth often resort to unsustainable activities like charcoal burning and mangrove cutting, worsening degradation and economic vulnerability. The elderly and PADs also struggle as resource depletion limits access to essentials. Households living adjacent to mangrove area are directly exposed to the impacts of deforestation, sedimentation, and pollution that reduce the ecological health of these habitats. Uncontrolled cutting of mangroves for wood fuel and construction materials, combined with the expansion of settlements and fish landing sites into sensitive areas, has increased human ecosystem interaction and heightened degradation pressure. The decline of mangrove cover has also reduced the natural buffering capacity against tidal surges and coastal erosion, exposing households and fishing infrastructure to greater physical and economic risks. In areas, where mangrove-dependent activities such as fishing and crab harvesting are central to livelihoods, the loss of mangrove resources directly threatens household income and food security.

3.4.7 Coral bleaching

Coral bleaching in Timboni has intensified markedly, shifting from sporadic, localized incidents in the late 1990s and early 2000s to frequent, widespread events since 2020. By 2025, bleaching has become a recurrent hazard, driven by rising sea surface temperatures and declining reef resilience. Repeated thermal stress has caused extensive coral mortality, reduced live coral cover, and altered reef composition, diminishing habitat complexity and biodiversity. Sediment runoff, nutrient pollution, and algal overgrowth have further hindered coral recovery, weakening the reef's ecological functions as a carbon sink, breeding ground, and natural barrier against erosion. The resulting degradation has lowered fisheries productivity, reduced tourism potential, and increased coastal vulnerability to storms and wave action. These impacts have severely affected small-scale fishers and tourism operators, with men in fisheries facing growing economic hardship and reduced adaptive capacity.

3.4.7.1 Exposure to Coral bleaching

Recurrent coral bleaching in Timboni has weakened reef structures, reduced live coral cover, and impaired ecological functionality, disrupting essential ecosystem services such as fish habitat provision, shoreline protection, and nutrient cycling. These cumulative impacts mark a shift from isolated thermal events to a persistent bleaching hazard driven by climate change, pollution, and unsustainable coastal practices.

3.4.7.2 Socio-Demographic Vulnerability to Sea Level Rise

Timboni's vulnerability to coral bleaching stems from its strong dependence on marine resources, gendered livelihood roles, and limited adaptive capacity. Most men rely on small-

scale artisanal fishing, while women engage in fish processing and trading, activities directly affected by coral degradation and declining fish stocks. This disruption has reduced household incomes, increased food insecurity and strained local economies. Limited access to credit, modern gear, and alternative livelihoods further constrains adaptation efforts. The loss of coral cover has also weakened natural coastal defenses, exposing settlements and infrastructure to wave action, erosion, and storm surges, while reduced marine productivity has intensified competition and conflict over dwindling resources. As men face declining fishing returns, women shoulder greater household and economic burdens, often resorting to unsustainable coping strategies. coral bleaching has become a multidimensional challenge in Timboni, undermining livelihoods, widening gender disparities, and eroding community resilience.

3.5 Classification of Exposure and Resilience through Livelihoods

The County's livelihood systems are deeply intertwined with its natural resource base, making them highly vulnerable to environmental, economic, and institutional shocks. Despite its diverse ecological and economic potential, spanning small-scale agriculture, marine and coastal activities, and informal trade, the county's resilience remains constrained by limited income diversity, low profit margins and persistent poverty. According to county data, Kilifi's overall poverty headcount stands at approximately 46.1%, well above the national average, highlighting widespread structural vulnerability. The livelihoods that sustain most households are predominantly informal and climate-sensitive, including farming, fishing, transport services, petty trade, and modest supplementary activities such as livestock keeping, poultry rearing, and palm wine tapping. Income stability across these sectors is increasingly undermined by droughts, market instability, and resource depletion, underscoring the need for a ward-level classification of exposure and resilience through a livelihood lens.

3.5.1 Livelihood Structure and Monthly Incomes

Livelihood systems across the County are diverse yet heavily dependent on natural ecosystems and informal market linkages. The findings reveal significant variation in household income across locations, reflecting differences in resource access, ecological settings, and exposure to environmental risks. Communities derive their livelihoods from multiple ecosystem goods and services provided by critical landscapes such as Arabuko-Sokoke Forest, Mwangea Hill, Kaya Godoma, and Kaya Kadzitsoni, which support a range of activities including agriculture and forest product harvesting among others. These ecosystems play an integral role in sustaining local economies through the regulation of microclimates, provision of raw materials, and maintenance of soil and water resources essential for production.

Across wards such as Dzikunze, Ndugumnani, Mwele, and Midugani, dominant livelihood activities include sand harvesting, charcoal burning, and motorbike transport, which constitute the upper-income tier, with typical monthly earnings of KES 48,000, 30,000, and 24,000 respectively. Sand harvesting has emerged as one of the most lucrative activities, particularly among men and youth, due to growing demand for construction materials in nearby towns. The activity provides immediate cash returns and supports a chain of loaders, transporters, and traders, making it a cornerstone of the informal local economy. Similarly, motorbike transport has become an indispensable service for community mobility, facilitating access to schools, health facilities, and markets. Households spend an estimated KES 200 per day on school transport alone, ensuring stable demand and consistent income for operators despite challenges such as poor road infrastructure and fluctuating fuel prices.

Charcoal burning, although economically important for many low-income households, presents a complex trade-off between short-term financial gain and long-term environmental degradation. The activity provides accessible income with minimal capital requirements, particularly for households living adjacent to forested landscapes. However, unregulated

production contributes to deforestation, habitat loss, and soil erosion, threatening the ecological integrity of surrounding ecosystems and undermining the sustainability of other nature-dependent livelihoods such as beekeeping, herbal medicine collection, and farming.

Lower-income livelihood options, such as subsistence agriculture, firewood collection, casual labour, small-scale trade, and water vending, generate modest and often irregular earnings ranging between KES 2,000 and 10,000 per month. These activities are primarily undertaken by women, youth, and vulnerable households with limited access to land, capital, and markets. While subsistence farming ensures household food security, it contributes minimally to cash income and remains highly vulnerable to erratic rainfall, pest infestations, and land degradation. Firewood collection, herbal sales, and casual labour depend heavily on seasonal demand and ecosystem availability, offering limited savings potential and low resilience to climatic and economic shocks.

In more inland communities such as Mbuyuni, Ondoni, Silala, and Vwevwesi, livelihoods are shaped by a mix of aggregate mining, livestock production, and small-scale agriculture. Mining stands out as the highest income source, averaging KES 50,000 per month, followed by agriculture (KES 20,000) and livestock rearing (KES 10,000–30,000). Despite its profitability, mining poses environmental challenges including soil erosion, flooding, and water contamination, underscoring the need for sustainable extraction practices to safeguard the ecosystems that underpin local livelihoods. Agriculture and livestock production continue to form the economic backbone of these areas, though productivity remains constrained by climatic variability, pest outbreaks, and market limitations.

3.5.2 Vulnerability of livelihood options

Livelihoods across Kilifi County, though central to household sustenance, are exposed to varying degrees of environmental and climate-related risk. The community's economic activities, ranging from sand harvesting, agriculture, and charcoal burning to small-scale trade, motorbike transport, and forest-based product collection, are all highly dependent on natural resources and sensitive to climatic fluctuations. This heavy reliance on fragile ecosystems, combined with limited diversification, weak social protection mechanisms, and persistent poverty, amplifies overall livelihood vulnerability and reduces adaptive capacity.

Sand harvesting, one of the most profitable activities, faces significant environmental and climatic challenges. Excessive extraction along seasonal rivers and near forested catchments disrupts groundwater recharge, accelerates soil erosion, and alters river morphology. During heavy rainfall or flooding, operations are halted, while prolonged droughts reduce sediment replenishment. Unsustainable harvesting practices also threaten biodiversity and hydrological balance, rendering this livelihood ecologically and economically unsustainable over time.

Charcoal burning, another key source of household income, depends heavily on forest resources drawn from areas around Arabuko-Sokoke and local Kaya forests. Forest degradation, coupled with regulatory restrictions, has reduced wood fuel availability, compelling producers to travel long distances or operate illegally. The practice itself accelerates deforestation, leading to soil degradation, increased surface temperatures, and reduced rainfall reliability, all of which undermine long-term livelihood security and intensify vulnerability to climate shocks.

Motorbike operations, while among the highest income-generating activities, are acutely sensitive to external shocks. Fluctuating fuel prices, maintenance costs, and deteriorating road conditions during heavy rains significantly affect profitability. Flooding and impassable rural roads frequently disrupt operations, while accident risks, lack of insurance, and breakdowns

expose riders to sudden financial losses. Despite their income potential, these factors make motorbike transport a high-return but high-risk occupation.
Agriculture, the most widespread livelihood activity, is directly impacted by climatic and ecological stressors. The predominance of rain-fed systems exposes crops to drought, erratic rainfall, and flooding, leading to low yields and frequent crop losses. Farmers also face pest outbreaks, declining soil fertility, and human–wildlife conflicts, particularly crop raids by monkeys, baboons, and elephants, which exacerbate food insecurity and erode household income. The inability to invest in improved seeds, fertilizers, and irrigation further limits resilience and adaptive capacity.

Livestock production faces similar constraints. Drought, disease outbreaks, and water scarcity reduce pasture availability and productivity, while floods and extreme weather events increase animal mortality. Livestock grazing near forested zones such as Arabuko-Sokoke also heightens exposure to predation and competition for feed. Limited access to veterinary services and fluctuating market prices further diminish the sector's resilience, leaving households vulnerable to repeated shocks.

Small-scale trade and informal services, which supplement household income and are dominated by women and youth, are equally fragile. Flooding, poor drainage, and reduced customer turnout during extreme weather disrupt local markets and diminish profitability. Seasonal droughts and economic downturns reduce local purchasing power, directly impacting traders' earnings. Similarly, forest-based activities such as herb collection, weaving, and beekeeping depend on the health and seasonal productivity of forest ecosystems. Overharvesting, drought, and irregular rainfall reduce both product quality and availability, while the burden of domestic chores and caregiving limits women's ability to adapt or diversify income streams.

Supplementary livelihoods, including firewood collection, water vending, and casual labour, remain highly climate-sensitive. Drought reduces water availability and increases competition for firewood, while floods disrupt transportation, damage infrastructure, and contaminate water sources. Casual labour opportunities decline sharply during extreme weather, especially when agricultural and construction activities are halted.

Underlying these vulnerabilities is the persistent issue of poverty. Most households operate within or just above subsistence thresholds, lacking savings, insurance, or formal financial access. Consequently, even minor external disturbances, such as a week of poor weather, a spike in fuel costs, or a household illness, can severely undermine economic stability. The combination of environmental stressors, income volatility, and limited institutional support perpetuates a cycle of vulnerability, constraining the county's capacity to build sustainable and climate-resilient livelihoods.

3.5.3 Overall Classification and Implications

The cumulative evidence positions Kilifi County within a high-exposure and low-to-moderate-resilience category, characterized by strong dependence on climate-sensitive and informal livelihood systems. Communities across the wards rely predominantly on subsistence agriculture, small-scale fishing, forest resource use, and informal trade, activities that are increasingly vulnerable to climatic shocks such as droughts, flooding, and ecosystem degradation. High poverty levels, coupled with entrenched gender disparities and extensive unpaid care and domestic work, continue to erode adaptive capacity and reinforce household vulnerability.

Building resilience across the County will require integrated and multi-sectoral interventions that stabilize incomes in high-risk sectors while diversifying livelihoods into higher-value and climate-resilient activities. Priority actions include strengthening access to microfinance and cooperative savings schemes, promoting climate-smart agriculture and sustainable fisheries, and expanding community participation in forest and mangrove conservation. Investments in ecosystem-based adaptation (EbA), such as the restoration and sustainable management of Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni, will enhance ecological integrity and buffer communities against climate-induced hazards.

Complementary improvements in rural infrastructure, including access roads, cold storage facilities, and irrigation systems, will improve market connectivity and reduce post-harvest losses. Reducing women's unpaid workload through labour-saving technologies, childcare support systems, and inclusive social protection mechanisms will further enhance adaptive capacity and gender equity. Embedding these measures within local development and climate action plans will help Kilifi County communities transition from reactive coping strategies toward proactive, sustainable, and equitable climate resilience (Source: PCVA, 2025).

3.6 Analyzing causes of vulnerability

3.6.1 Classification of Severity of Vulnerabilities and Their Causes

Kilifi County exhibit differentiated levels of vulnerability to climate hazards, driven by ecological setting, livelihood dependence, and adaptive capacity. Vulnerability patterns across sampled villages demonstrate a clear spatial and socio-economic gradient, where communities situated near coastal and forested ecosystems face heightened exposure to climatic and environmental stressors.

In Sokoke Ward, villages such as Mwele and Ndugumnani are classified as highly vulnerable due to their heavy reliance on resource-dependent livelihoods, particularly sand harvesting, charcoal burning, and rain-fed agriculture, that are acutely sensitive to climatic shocks. Recurrent droughts, flooding, and land degradation reduce productivity and increase livelihood uncertainty. In contrast, Dzikudze and Midugani display moderate vulnerability, supported by a relatively higher degree of livelihood diversification, though income sources remain sensitive to market fluctuations and environmental variability.

In Mnarani Ward, high vulnerability is observed in Vwevwesi and Ondoni, where communities depend on forest resources from Kaya Chonyi for fuelwood, timber, and herbal medicine. These villages face recurrent drought, human–wildlife conflict, and depletion of forest ecosystems. Mbuyuni and Silala are moderately vulnerable, with livelihoods threatened by drought, pest infestations, and limited access to water. Agricultural households face recurrent crop failures and reduced food security due to erratic rainfall and declining soil fertility (Source: PCVA, 2025).

In Mwarakaya Ward, coastal and near-coastal settlements such as Takaungu, Vuma, Vijiweni, Mikwajuni, and Timboni exhibit high vulnerability due to their dependence on marine and coastal ecosystems, coupled with exposure to saline intrusion, flooding, and sea-level rise. Midzi Mitsano is moderately vulnerable, where drought and pest infestation periodically disrupt agricultural production and water availability. These stressors are compounded by limited access to climate information, weak infrastructure, and inadequate livelihood diversification.

Across all three wards, the severity of vulnerability is amplified by chronic poverty, limited access to finance and technology, and gendered unpaid labour that constrain adaptive capacity. Women remain disproportionately affected due to their concentration in low-return and

resource-dependent activities, restricted access to credit and training, and the burden of unpaid domestic work, which intensifies during climate crises.

This classification underscores the multidimensional nature of vulnerability in Kilifi County, where ecological exposure, socio-economic status, and gender dynamics intersect to determine overall risk. Addressing these vulnerabilities requires strengthening community-based adaptation, enhancing access to finance and infrastructure, and mainstreaming gender-responsive strategies to build resilience across all wards (Source: PCVA, 2025).

3.6.2 Identification of Unsafe Conditions

Unsafe conditions across Sokoke, Mnarani, and Mwarakaya Wards represent the immediate environmental and socio-economic contexts that heighten exposure to climate-related hazards and weaken household safety in Kilifi County. In Sokoke Ward, villages such as Mwele, Ndugumnani, and Dzikudze face recurrent human–wildlife conflicts and environmental degradation arising from unsustainable activities such as sand harvesting, herb collection, and firewood gathering. These practices deplete forest cover within the Arabuko Sokoke ecosystem, reducing its capacity to regulate floods, recharge groundwater, and buffer communities from droughts. Poor road access, weak drainage systems, and limited market infrastructure further restrict mobility and economic activity during extreme weather events, compounding livelihood disruptions and delaying recovery.

In Mnarani Ward, unsafe conditions are driven by settlements situated along riverine and forested zones, particularly in Ondoni and Vwevwesi, where recurrent flooding, drought-induced water scarcity, and soil erosion threaten homes and farms. Inadequate drainage and encroachment into riparian areas exacerbate flood risks, while informal housing in low-lying or unstable areas increases exposure to structural damage during heavy rains. Weak enforcement of land-use regulations and lack of disaster preparedness mechanisms have resulted in communities living and working in hazardous environments, where infrastructure damage frequently interrupts access to education, healthcare, and markets (Source: PCVA, 2025).

In Mwarakaya Ward, unsafe conditions are most pronounced in coastal settlements such as Takaungu, Vuma, Vijiweni, and Mikwajuni, which experience tidal flooding, saline intrusion, and shoreline erosion due to their proximity to the ocean and degradation of protective mangrove forests. Inadequate solid waste management and poor drainage have intensified flood impacts, while the absence of safe fish landing sites, raised storage facilities, and weather-resistant shelters increases occupational and residential risk. From a gender and social perspective, women, youth, and Persons Abled Differently (PADs) bear the brunt of these unsafe conditions. Women and girls travel long distances through degraded and unsafe landscapes to collect water and firewood, while households headed by widows and the elderly face mobility and resource constraints that limit adaptation. Collectively, these conditions illustrate how environmental degradation, infrastructural deficits, and gender inequities converge to amplify vulnerability, underscoring the need for ecosystem-based adaptation, resilient infrastructure, and inclusive planning to safeguard livelihoods and enhance resilience (Source: PCVA, 2025).

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures across Sokoke, Mnarani, and Mwarakaya Wards represent the intermediate socio-economic and institutional processes that translate underlying weaknesses into direct unsafe conditions, heightening vulnerability to climate-related shocks in Kilifi County. These pressures are largely driven by rapid population growth, land-use conversion, overexploitation of natural resources, and weak governance frameworks. In Sokoke Ward, unregulated settlement expansion and unsustainable land clearing for agriculture have contributed to the

severe degradation of the Mwangea Hills ecosystem, a vital water catchment area whose forest cover has been almost entirely lost. Prolonged droughts have forced communities to convert forested lands into farms, while limited restoration initiatives and inadequate enforcement of conservation by-laws have allowed degradation to persist. Similarly, in Mnarani Ward, mangrove deforestation in Takaungu, intensified by both human encroachment and infrastructural development, has led to sedimentation, coastal erosion, and reduced fish stocks, directly undermining marine-based livelihoods.

In Mwarakaya Ward, degradation of Kaya Chonyi Forest, one of the ward's key ecological and cultural assets, illustrates how dynamic pressures translate into ecological and social vulnerability. Encroachment for settlement and cultivation, unsustainable grazing, and unregulated harvesting of timber and firewood have diminished vegetation cover and biodiversity, disrupting vital ecosystem functions such as soil stabilization, microclimate regulation, and water retention. High population growth and unemployment have intensified forest exploitation, while unresolved land disputes and weak local cohesion continue to strain conservation efforts. The degradation of forest and mangrove ecosystems across the three wards not only weakens their role as natural buffers against droughts, floods, and coastal hazards but also amplifies the vulnerability of households reliant on these ecosystems for subsistence and income.

From an institutional and socio-economic perspective, limited coordination between county departments, community-based organizations, and national agencies constrains the implementation of effective resource governance and restoration programs. Enforcement of by-laws remains inconsistent, and early warning systems for droughts and floods are poorly developed, leaving communities reactive rather than prepared. Socially, gender inequalities and the exclusion of women, youth, and marginalized groups from decision-making forums, particularly in forest and fisheries management, reduce inclusivity and ownership of adaptation strategies. Collectively, these interlinked pressures reinforce a cycle of environmental degradation and livelihood insecurity, underscoring the need for integrated governance, sustainable resource use, and inclusive adaptation planning across the County (Source: PCVA, 2025).

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability across Kilifi County arise from interlinked socio-economic, environmental, and institutional deficiencies that collectively erode community resilience and adaptive capacity. Structural poverty remains pervasive, characterized by limited access to education, technology, financial services, and diversified livelihoods. According to the Participatory Climate Vulnerability Assessment (PCVA, 2025), household sizes average between 7 and 9 persons per household, indicating high dependency ratios that strain limited natural and financial resources. Most households depend on climate-sensitive and ecosystem-based livelihoods such as small-scale agriculture, fishing, charcoal production, and herbal medicine trade. This dependence not only sustains poverty cycles but also heightens exposure to environmental and economic shocks, as resource degradation directly undermines household income stability and food security.

Environmental degradation represents another critical underlying cause, driven by unsustainable exploitation of natural resources, weak land-use management, and recurrent droughts. The degradation of Mwangea Hills, Kaya Chonyi Forest, and the Takaungu mangrove ecosystem illustrates the cumulative effects of deforestation, encroachment, and overharvesting of natural vegetation. These ecosystems, once vital buffers against floods, droughts, and coastal erosion, have lost their protective capacity, exposing communities to increased climatic stress and declining productivity. Soil erosion, saline intrusion, and loss of biodiversity have further

compounded these impacts, while the scarcity of arable land and declining marine productivity push communities toward increasingly unsustainable coping strategies such as sand harvesting and charcoal burning.

From an institutional and social perspective, weak governance structures, overlapping mandates, and limited coordination between the County Government of Kilifi, Ward Climate Change Planning Committees (WCCPCs), and national agencies have hindered the implementation of adaptive interventions. Limited financial resources, inadequate enforcement of environmental by-laws, and absence of comprehensive early warning and information dissemination systems have resulted in delayed responses to climatic hazards, shifting the focus from preparedness to crisis management. Moreover, gender inequality remains a significant underlying determinant of vulnerability. Patriarchal norms and customary land tenure systems restrict women's ownership and control over productive assets, reducing their participation in decision-making and access to credit and climate-smart technologies. As a result, women remain concentrated in low-return and climate-sensitive livelihood activities, reinforcing their economic marginalization and constraining the overall adaptive capacity of the household and community (Source: PCVA, 2025).

3.6.5 Prioritization of Causes

The prioritization of the underlying causes of vulnerability across Kilifi County was based on participatory ranking during the Participatory Climate Vulnerability Assessment (PCVA, 2025), using criteria such as frequency, severity, and reversibility of impacts. The highest priority causes identified were poverty, environmental degradation, and weak institutional support systems. Persistent poverty restricts households from making adaptive investments and perpetuates dependence on climate-sensitive livelihoods such as small-scale farming, fishing, charcoal burning, and harvesting of forest products. Environmental degradation, including deforestation in Arabuko Sokoke and Mwangea Hills, destruction of mangrove forests in Takaungu, and encroachment in Kaya Chonyi, has weakened the natural ecosystems' capacity to regenerate and buffer communities against climate shocks. Furthermore, weak institutional mechanisms, characterized by fragmented coordination, inadequate financial support, and limited enforcement of environmental by-laws, delay effective response and hinder the implementation of ward-level adaptation and restoration initiatives.

Medium-priority causes identified include gender inequality, population pressure, and limited access to climate information, which significantly shape vulnerability dynamics across the wards. Addressing these priority causes requires a multi-sectoral and inclusive approach that strengthens governance frameworks, promotes livelihood diversification, and ensures gender-responsive climate action. Priority interventions include the restoration of degraded terrestrial, forest, and mangrove ecosystems; expansion of microcredit access for women's enterprises; capacity building for local climate governance institutions; and establishment of community-based early warning and disaster preparedness systems. Collectively, these measures will enhance adaptive capacity, reduce long-term vulnerability, and promote sustainable and equitable resilience across all villages in the County (Source: PCVA, 2025).

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards

Gender and age critically determine exposure, sensitivity, and adaptive capacity across population groups in Kilifi County, where climate hazards; including drought, floods, pests and diseases, environmental degradation, sea level rise, strong winds, human–wildlife conflicts, and coral bleaching, interact with deeply rooted socio-cultural inequalities and poverty dynamics (Table 2). These hazards disproportionately affect women, girls, youth, elderly persons, and Persons Abled Differently (PADs), whose vulnerability is shaped by unequal access to resources, restricted mobility, gendered labour roles, and limited influence over household and community-level decision-making (UN Women, 2022; IPCC, 2022).

A central driver of vulnerability among women and girls in Kilifi is the heavy and undervalued burden of unpaid domestic and care work, including childcare, cooking, firewood collection, water-fetching, elder care, cleaning, laundry, and subsistence farming. During Participatory Climate Vulnerability Assessments (PCVAs), this labour was valued at KES 800–1,500 per day, yet remains invisible in formal economic systems. Nationally, women perform over 70% of unpaid care work (KNBS, 2023; UNDP, 2022), and climate hazards intensify this workload by increasing the time required to secure water, food, healthcare, and fuelwood. This "time poverty" constrains women's participation in income-generating activities, leadership spaces, capacity-building training, and climate adaptation initiatives (ILO, 2018; UN Women, 2023).

Drought is one of Kilifi's most extensive hazards, exacerbating water scarcity and reducing livestock and crop productivity. Women and girls walk longer distances, sometimes 5 kilometres or more, to collect water and firewood, exposing them to fatigue, dehydration, musculoskeletal strain, and protection risks including harassment and assault. Female-headed households, common due to labour migration and widowhood, experience severe impacts due to limited financial resources and labour support. Men face income losses from reduced yields, livestock mortality, and declining forest resources, causing psychological stress. Youth (18–35) lose casual labour opportunities and apprenticeships, increasing idleness and risky coping behaviours (UNFPA, 2022). Elderly persons and PADs struggle to access water and food due to mobility constraints, increasing their dependency (WHO, 2021).

Floods destroy homes, early childhood education centres, roads, and markets, directly affecting women's livelihoods in informal trading, retail, and food vending. Women often care for injured or displaced relatives and manage contaminated water sources, intensifying stress and workload. Market closures reduce women's income significantly. Men lose income from fishing, transport, and casual labour due to dangerous water conditions and damaged infrastructure. Youth experience school closures, loss of livelihood opportunities, and heightened vulnerability to exploitation. Elderly individuals and PADs face evacuation challenges and high risk of neglect in congested temporary shelters (WHO, 2020).

Pests and crop/livestock diseases reduce agricultural yields, affecting women who dominate subsistence farming, poultry-keeping, and goat-rearing. Limited access to extension services, certified seeds, and pest control technologies reduces women's adaptive capacity (FAO, 2021). Men face losses in livestock and cash crops, while youth lose agricultural casual labour opportunities. Elderly persons and PADs experience heightened food insecurity due to reduced household harvests.

HWC is significant in Kilifi, especially near Arabuko-Sokoke Forest and coastal zones bordering creek systems. Women and girls limit their movements due to fear of elephants, baboons, hyenas, and crocodiles, affecting access to water, firewood, markets, and social networks. Women also struggle to navigate compensation systems due to limited documentation and restricted decision-making power. Men face crop and livestock losses and risk injury when

guarding farms. Youth assisting with farm protection face similar risks. Elderly persons and PADs are particularly vulnerable because of slower mobility and reduced ability to escape danger.

Environmental degradation in Kilifi County has intensified due to uncontrolled mining activities by Mombasa Cement and other quarrying companies, coupled with extensive deforestation for charcoal production around the Mwangea Hills. These practices have accelerated soil erosion, depleted groundwater, destabilised ecosystems, and reduced forest and mangrove cover, directly undermining natural resource availability for households. Women are disproportionately affected as they must walk longer distances to access firewood, water, and other non-timber forest products, further increasing their already heavy unpaid care workload. Men experience declining agricultural and fishing productivity as degraded landscapes disrupt catchment health, reduce streamflow, and damage breeding habitats. Youth lose opportunities in nature-based enterprises such as beekeeping, ecotourism, and seedling production, while elderly persons and Persons Abled Differently (PADs) face heightened dependency due to mobility constraints and reduced access to essential resources. Strong winds and storms cause house destruction, loss of temporary structures, and injuries. Women bear increased caregiving for injured family members and securing household items. Men, particularly fishermen, face high risks of capsizing and gear loss. Youth face injury from falling structures, while elderly persons and PADs have difficulty accessing safe shelters.

Coral bleaching is a major climate hazard along the Mnarani ward coastline, where rising sea temperatures have led to significant reef mortality. The degradation of coral reefs disrupts marine food webs, reduces fish stocks, and undermines tourism potential. Women engaged in gleaning, seaweed farming, nearshore fisheries, and coastal micro-enterprises experience income losses as reef ecosystems decline. Bleaching reduces the availability of shellfish, octopus, and juvenile fish, key sources of household nutrition. Men in artisanal fishing spend more on fuel and face greater safety risks as they travel further offshore. Youth lose jobs in snorkeling, guiding, boat operations, and other reef-based tourism services. Elderly persons and PADs face increased dependency because households lose marine-based income and protein sources.

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Kilifi County. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities , where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to productive assets, and participation in ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience buildin

Table 2: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Female-headed households bear highest burden of water, food, and fuelwood provision • Walk longer distances to depleted water sources → fatigue, dehydration, back strain • Reduced time for childcare, rest, and income generation • Increased psychological stress due to food shortages	• Livelihood insecurity from reduced crop yields, livestock productivity, and forest resources • Increased financial pressure and stress • Reduced availability of agricultural labour opportunities	• Reduced productivity and income from casual labour losses • High idleness → increased risk of crime, drugs, and unsafe coping strategies • Higher incidence of early marriages and pregnancies among girls	• Mobility limitations restrict access to water and food • Increased susceptibility to heat stress, dehydration, and hunger • Dependence on caregivers worsens during drought periods	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Increased caregiving burdens for sick or displaced family members • Loss of income from informal trade due to market inaccessibility	• Loss of fishing, transport, and casual labour opportunities • Psychological stress from inability to provide for families	• School closures disrupt learning and training • Loss of casual labour -increased idleness, crime, and drug use • Girls particularly vulnerable to early	• High susceptibility to waterborne diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter,	• Severe mobility barriers during evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services

Women	Men	Youth	Elderly	PADs
• Contaminated wells force longer distances to find safe water • Elevated psychosocial stress	• Damage to tools and assets disrupts livelihoods	marriage and pregnancy	food, and medicine	
Pests and diseases				
• High exposure due to their role in managing subsistence farms, small livestock, and seaweed farming • Reduced food security and income from harvest losses • Limited access to pest-control technologies and extension services	• Income loss from crop and livestock declines • Increased stress due to reduced labour demand and food shortages • Higher pressure to diversify livelihoods	• Loss of agricultural casual labour opportunities • Increased unemployment and financial vulnerability • Greater exposure to negative coping behaviours	• Increased dependence for food and care during outbreaks • Reduced capacity to protect crops and livestock • Nutritional deficits worsen chronic illnesses	• High dependency on family for food and adaptive support • Limited access to adapted technologies and extension services • Increased vulnerability from household food shortages
Human wildlife conflicts				
• Restricted mobility due to fear of wildlife attacks • Loss of crops and livestock increases household food insecurity	• High risk of injury when guarding farms and livestock, often at night • Economic strain from repeated	• Increased risk when helping to guard farms • Reduced productivity and fewer livelihood opportunities	• Very limited capacity to respond to wildlife encounters • Increased dependence after	• Inability to flee or protect themselves during wildlife incidents • Extremely high dependence on caregivers for

Women	Men	Youth	Elderly	PADs
• Limited access to compensation processes	crop and livestock losses • Pressure to seek alternative income sources	• Higher vulnerability to migration and exploitation	livestock or crop losses • Heightened food insecurity	safety and recovery • High vulnerability during compensation or reporting processes
Environmental Degradation				
• Reduced access to firewood, mangroves, herbs, and NTFPs • Walk longer distances (5+ km) for resources leading to exhaustion • Less time for childcare and economic activities	• Declining fish stocks increase offshore travel distance and fuel costs • Higher fishing risks • Reduced forest resource availability increases livelihood strain	• Fewer forest-based livelihood options (beekeeping, nurseries, eco-tourism) • Increased involvement in illegal extraction when opportunities shrink • Rising idleness and economic vulnerability	• Mobility challenges increase dependency during resource scarcity • Physical strain from long distances to collect resources • Heightened vulnerability to malnutrition	• Limited access to degraded resource areas • Exclusion from resource-based livelihood opportunities • Increased reliance on caregivers
Strong winds				
• Increased caregiving demand after home damage • Higher risk securing household assets during storms	• High danger at sea and gear damage • Increased risk of capsizing during strong winds	• High injury risk from collapsing structures or debris • Limited ability to seek safe shelter	• Difficulty accessing safe shelter quickly • High injury and exposure risk	• Severe challenges evacuating safely • High exposure to falling debris and structural collapse
Coral bleaching				

Women	Men	Youth	Elderly	PADs
• Reduced income from seaweed farming • Decline in tourism affects women in micro-enterprises (kiosks, vending) • Loss of nearshore ecosystems reduces access to marine resources	• Declining fish stocks reduce artisanal fishing catch • Increased fuel and operational costs as fishing shifts offshore • Higher safety risks at sea	• Reduced employment in snorkelling, tour guiding, and boat operations • Lower household income increases vulnerability to risky coping strategies • Decline in coastal tourism limits livelihood pathways	• Reduced access to fish lowers food security • Limited mobility restricts ability to shift livelihood practices • Higher dependency on family support	• Lost opportunities in tourism and fisheries-based livelihoods • Increased dependency as households face reduced marine resources • Higher risk of exclusion from alternative livelihood training

4.1 Children's Vulnerability to Climate Hazards and Key Mitigation Measures

Children in Kilifi County represent one of the most climate-vulnerable population groups due to their reliance on caregivers, rapid developmental needs, and limited capacity to anticipate or respond to hazards (UNICEF, 2022). Their vulnerability is compounded by malnutrition, poverty, limited WASH coverage, and the impacts of displacement or school disruption.

During droughts, children face increased risks of acute malnutrition, stunting, dehydration, and waterborne diseases as households turn to unsafe water sources. Many children, especially girls are tasked with water collection, exposing them to heat stress, exhaustion, and protection risks that hinder school participation (UNICEF, 2022). Floods destroy homes, schools, and WASH facilities, causing drowning risks, respiratory infections, diarrhoeal diseases, and cholera outbreaks. Learning interruptions occur when schools are converted into evacuation shelters.

Environmental degradation and coral bleaching threaten child nutrition by reducing fish availability, especially in coastal households where fish forms a major protein source. Human–wildlife conflict poses risks during daily routines such as grazing livestock, firewood collection, or market errands. Sea level rise contributes to recurrent displacement, disrupting social networks and creating psychosocial trauma for children, including anxiety, nightmares, academic regression, and behavioural issues (UNICEF, 2021).

Poverty and gender norms deepen childhood vulnerability. Families often resort to harmful coping strategies such as early marriage, withdrawal of children from school, child labour, or transactional activities among adolescents (UNFPA, 2022). These harmful practices violate child rights and perpetuate intergenerational poverty and climate vulnerability.

Protective measures require child-centred, multi-sectoral climate adaptation. Strengthening early warning systems in schools and communities ensures timely action. Enhancing nutrition surveillance and early action protocols enables rapid detection and treatment of malnutrition. Climate-resilient WASH infrastructure, including elevated latrines, protected boreholes, and rainwater harvesting systems, reduces disease outbreaks during floods and droughts. Climate-proof education infrastructure with storm-resistant roofs, elevated floors, and shaded play areas safeguards learning continuity (UNESCO, 2021).

Provision of psychosocial support, child-friendly spaces, trauma counselling, and family-tracing services is critical during displacement. Expanded social protection, including school feeding, bursaries, and child grants, helps households avoid harmful coping strategies during climate shocks (UNDP, 2022). Strengthening community-based child protection structures ensures that teachers, CHVs, and local leaders safeguard children's welfare before, during, and after climate hazards.

5 Chapter 5: Adaptation Action Plan

5.1 Strategic Adaptation Objectives and Thematic Priorities

To effectively reduce climate risks and enhance resilience in Kilifi, the CCAAP adopts a set of strategic adaptation objectives organized around key thematic priority areas. These priorities align with the major hazards and vulnerabilities identified, and with national and county strategy frameworks (National Adaptation Plan (NAP), National Climate Change Action Plan (NCCAP) and County Integrated Development Plan (CIDP). Each thematic area encompasses specific goals and actions, described below, with an emphasis on sustainability, community ownership, and co-benefits (such as livelihoods and ecosystem health). Cross-cutting issues like capacity building, gender and social inclusion, and climate information services are mainstreamed throughout all themes.

Water Security and Climate-Resilient Agriculture: Ensuring reliable water access and promoting climate-resilient agriculture constitute core components of this adaptation action plan. Key interventions constitute of the construction of boreholes with solar-powered pumps, rehabilitation and desilting of water pans, and the enhancement of household rainwater harvesting through tanks and gutter systems, collectively aimed at enhancing year-round water availability. Complementing these infrastructure measures, the plan emphasizes on capacity-building in climate-smart agriculture, encompassing training on drought-tolerant crop varieties, adoption of resilient livestock breeds, and implementation of efficient irrigation systems such as drip irrigation. By improving year-round water availability, the plan not only reduces drought impacts but also yields co-benefits in health (clean water access) and gender equity, women and girls benefit significantly from shorter water-fetching distances and times. Overall, strengthening water security is foundational to adaptation, touching on livelihoods, health, and conflict reduction

Food Security and Livelihood Resilience: Food security and resilient livelihoods are supported through interventions targeting pests, diseases, and agricultural productivity. The adaptation action plans include establishing farmer field schools and demonstration plots to facilitate practical learning on pest surveillance, safe pesticide use, biological control methods, and climate advisories for planting timing, dissemination of drought-tolerant and pest-resistant crop varieties to increase yield, and the provision of improved livestock breeds to reduce increase productivity. Furthermore, regular veterinary outreach and livestock vaccination campaigns ensure healthy herds, while training in crop and livestock insurance strengthens risk management capacity and minimizes economic losses. Additionally, the plan advocates for livelihood diversification through adoption of alternative livelihoods such as beekeeping and mariculture, thereby strengthening household resilience, particularly among marginalized communities, by providing alternative sources of income that are less susceptible to climate variability

Ecosystems and Natural Resource restoration: Ecosystem management forms a central pillar of this adaptation plan, recognizing that healthy and functional ecosystems are essential for climate resilience and sustainable livelihoods. Restoration efforts will prioritize degraded mangrove and terrestrial forest ecosystems, which serve as natural buffers against climate hazards. These ecosystems regulate hydrological cycles, stabilize soils, and attenuate storm surges, thereby reducing community vulnerability to climate impacts. Key interventions include the rehabilitation of degraded forests and watersheds through tree planting and community-led afforestation on denuded lands, coupled with the conservation of critical coastal and marine ecosystems. These actions will be complemented by soil and water conservation measures such as terracing, establishment of vegetative buffers, and erosion control structures to enhance landscape stability and productivity. In addition, the promotion of energy-efficient cookstoves aims to reduce dependence on charcoal and firewood, thereby curbing deforestation and

pressure on natural forests. Community awareness programs and enforcement of environmental regulations will further ensure the sustainable management of forest and coastal resources. Collectively, these measures enhance ecological integrity, strengthen carbon sequestration potential, and secure vital ecosystem services that underpin livelihoods, while advancing long-term climate resilience and sustainable development.

Enhanced Resilience to Extreme Weather Events: This action plan prioritizes strategic interventions to mitigate the impacts of extreme weather events, including floods, droughts, and high wind episodes, that pose significant risks to livelihoods and community wellbeing. To enhance resilience of critical infrastructure and settlements to these hazards, the plan prioritizes climate-proofing investments in transport, drainage, and disaster preparedness systems. Core actions include constructing or elevating critical bridges and culverts at flood-prone crossings to maintain all-weather connectivity and improving road drainage systems, particularly along rural access roads to prevent erosion, washouts, and community isolation during heavy rains. Floods management measures will involve reinforcing riverbanks, construction of dams to control floodwaters and desilting drainage channels to restore natural flow capacity. In parallel, the county will strengthen community Early Warning Systems (EWS) by integrating Kenya Meteorological Department (KMD) forecasts with local volunteer networks to ensure timely dissemination of alerts through SMS and community communication channels. Collectively, these measures will build the resilience of both the built and natural environments, reducing disaster losses and safeguarding lives, livelihoods, and critical infrastructure.

Gender mainstreaming and community governance : To ensure that climate adaptation is inclusive, equitable, and institutionally grounded, the plan integrates gender and social inclusion as core guiding principles. Gender mainstreaming will be systematically integrated into all programs through targeted capacity-building and inclusive participation strategies, led by the County Gender & Social Services Department, Ward Climate Change Planning Committees (WCCPC), Gender Desk, NGOs, and Ward Administration. It shall constitute meaningful participation and equitable benefit-sharing among women, men, youth, and Persons Abled Differently (PADs) across all adaptation initiatives. Recognizing that household decision-making and resource governance has historically been male dominated, the plan deliberately promotes women's leadership and representation within Ward Climate Change Planning Committee (WCCPC), Water Resource User Associations (WRUAs) and Community Forest Association (CFA), enabling their active involvement in local planning and resource allocation processes. Further, the plan will institutionalize gender-responsive budgeting to ensure equitable resource distribution and strengthen the capacity of women and youth through targeted training in leadership, advocacy, and entrepreneurship. These measures aim to enhance agency, improve access to decision-making spaces, and embed gender justice at the core of local adaptation governance thereby ensuring that climate actions are socially just, participatory, and sustainable.

Cross-cutting support: All the above thematic actions are supported by cross-cutting strategies in capacity building, climate information services, disaster preparedness, and inclusion. Public awareness and education campaigns will be carried out to inform citizens about climate risks and adaptation practices (e.g. water conservation). The plan will strengthen early warning systems for multiple hazards (drought, floods) and ensure timely communication to communities (through radios, SMS, community barazas). Continuous capacity building of local institutions (WCCPCs, CBOs) and county staff is planned so that they can effectively plan, implement, and monitor adaptation initiatives.

5.2 Adaptation Actions and Implementation Plan

The adaptation actions identified across Kilifi County were largely uniform across the wards, reflecting similar climate risks and community priorities. Based on findings from the

Participatory Climate Vulnerability Assessments (PCVAs), ward adaptation plans, and strategic priorities, a comprehensive framework of adaptation interventions to be implemented between 2025 and 2030 has been developed (Table 3). These actions respond to eight priority climate hazards; drought, floods, pests and diseases, strong winds, Environmental degradation, sea level rise, coral bleaching and human–wildlife conflicts and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kilifi County Climate Change Act (2021), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

Under Thematic Area One, Water Security and Management, the adaptation actions prioritise immediate enhancement of household and institutional water access across Mnarani, Mwarakaya and Sokoke Wards. Short-term actions focus on expanding rainwater harvesting through the installation of gutters and household-level water tanks in villages such as Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni and Vuma in Mnarani; Mbuyuni, Ondoni and Vwevwesi in Mwarakaya; and Dzikunze, Midugani, Mwele and Ndugumnani in Sokoke. These measures reduce the time women and girls spend fetching water and strengthen hygiene practices during drought periods. Complementary actions include fencing springs in Vwevwesi to safeguard water quality, constructing a sand dam along River Mwabanda in Ondoni, and sensitising women and youth on water treatment to improve household health outcomes. Medium-term measures emphasise rainwater harvesting in schools through installation of high-capacity 10 m^3 tanks, six schools in Mnarani, four in Mwarakaya and four in Sokoke, ensuring reliable water supply for children while reducing absenteeism linked to water scarcity. Strategic installation of solar-powered water pumps where viable embankments exist further aims to reduce the drudgery of water collection for women, especially in Takaungu, Vijiweni, Vuma, Ondoni, Vwevwesi and Midugani. Long-term actions include drilling and solarising new boreholes across the three wards, constructing water pans in all identified villages, and enhancing surface water storage and recharge through additional sand dams. Implementation will rely on Kilifi County Government, KIMAWASCO, CHPs, NGOs and private-sector partners, with budgets ranging from KES 50,000 per household tank, KES 400,000 per school tank, KES 5,000,000 per borehole, KES 3,000,000 per water pan and KES 1,000,000 per sand dam.

Under Thematic Area Two, Food Security, Ecosystem Restoration and Livelihoods, short-term actions aim at strengthening climate-resilient agriculture, biodiversity restoration and women- and youth-led agribusiness systems. Communities across the three wards will be trained and supported on Resilience Design techniques, soil fertility enhancement, Zai pits, water retention, and farm layout optimization, to reduce production costs through locally available inputs. Women and youth will also receive skills on Climate Smart Agriculture (CSA), covering drought-tolerant crops such as cassava, sorghum, katumani maize, sunflower and groundnuts, alongside training on silage and hay-making to stabilize livestock feed supplies. The programme also promotes cassava value addition, organic briquette making, and household-level waste management, strengthening both food security and environmental stewardship. Medium-term actions build on early gains by supporting lead farmers to establish fully functional resilient-design demonstration farms and providing certified seed varieties, agro-inputs and extension services to trained groups. Central waste collection bins in shopping centres, regulation of sand harvesting, and the establishment of fodder seed banks and improved livestock breeds enhance long-term ecosystem and livelihood resilience. Long-term actions involve establishing Farmer Field Schools in each ward; rehabilitating degraded hotspots such as Mwangea hills and Kaya Kadzistoni in Sokoke and Kaya Chonyi in Mwarakaya; implementing mangrove and coral reef

restoration in Takaungu Creek and Timboni; and constructing terraces, gabions and check dams to address erosion along rivers Mwanzae and Mwabanda. Key partners include Kilifi County Government, KARLO, KFS, KEFRI, CFAs, NEMA, WRA, NGOs and private entities. Budgets range from KES 10,000 per farmer per year, KES 75,000 per nursery, KES 150,000 per acre for land rehabilitation, KES 450,000 per acre of mangrove restoration and KES 3,000,000 for coral reef restoration, among other costed actions.

Under Thematic Area Three, Diversified Livelihoods and Social Protection, the short-term focus is on supporting women, youth and PADs to diversify income through beekeeping, poultry farming, basketry, ecotourism and mariculture ventures in Mnarani, Mwarakaya and Sokoke. Training enables entry into viable sectors such as milkfish, mud crab, catfish and tilapia aquaculture, alongside enhanced small-scale enterprise development in Sokoke. Provision of relief food and water during drought remains a critical safety net, especially for highly vulnerable households. Medium-term actions emphasise strengthening productive capacities through the supply of modern beehives, egg incubators, fishing boats, cages, fingerlings and fish feed pellet mills, as well as supporting blue carbon project registration in Takaungu Creek consistent with the Climate Change (Carbon Markets) Regulations, 2024. Long-term actions include constructing a cultural centre and nature trail at Kaya Chonyi to boost eco-tourism and promoting VSLA investments in green bonds to enhance financial resilience. Implementation by KeFS, NMK, KMFRI, BMUs, County Government and NGOs is supported by budgets including KES 7,000 per beehive, KES 100,000 per incubator, KES 1,000,000 per cage, KES 2,000,000 per boat, and KES 50,000 per household enterprise.

Under Thematic Area Four, Community Infrastructure and Disaster Management, short-term actions focus on training community emergency response teams, planting windbreaks in Vuma and Ndugumnani, and designating disaster assembly points and temporary camps. These interventions strengthen preparedness for floods, strong winds and drought across all three wards. In the medium-term, youth will participate in food-for-work programmes to restore drainage channels, clean culverts, construct gabions and plant trees, thereby enhancing both early recovery and community stewardship. Long-term actions include constructing raised footbridges and culverts in flood-prone villages such as Ondoni, Silala and Mbuyuni; improving drainage infrastructure in Timboni; equipping emergency response teams; operationalising multi-hazard early warning systems; and constructing new health facilities in Vijiweni, Vuma, Silala, Vwevwesi and Mwele. Implementation partners include Kilifi County Government, KERRA, KMD, NDMA, Kenya Red Cross, Ministry of Interior and private entities, with budgets such as KES 840,000 for ward-level EWS and BOQs for infrastructure works.

Under Thematic Area Five, Human–Wildlife Conflict Management, early interventions include sensitising communities in Ondoni, Vwevwesi, Dzikunze, Mwele, Midugani and Ndugumnani on conflict hotspots, safe behaviours and compensation procedures. Medium-term actions promote chili cultivation and apiculture along forest–farmland boundaries as nature-based deterrents to elephants and primates, thereby protecting farms without harmful retaliation. Long-term actions include installing a solar-powered electric fence around Kaya Chonyi to protect both cultural heritage and surrounding communities. KWS, Kilifi County Government and NGOs will lead implementation, with costs of KES 7,000,000 for fencing and statutory compensation rates for human injury or property loss.

Under Thematic Area Six, Gender and Social Inclusion, the programme emphasises anticipatory GBV actions related to drought, floods and strong winds. Short-term priorities include targeted sensitisation, strengthening women's rights platforms such as VSLAs, supporting PADs with assistive infrastructure, and holding youth boot camps on drugs, reproductive health and behaviour change. Medium-term actions focus on strengthening gender-responsive governance,

improving GBV response mechanisms, including transfer vehicles, specialized first-aid kits and rape kits, and promoting vaccination campaigns for children and youth exposed to climate-related disease risks. Long-term actions include establishing a GBV recovery center at county level and constructing a drug rehabilitation center for youth. The actions will be supported by the County Government, National Police Service, NGOs, civil society and private sector, with budgets including KES 1,000,000 for consultancy support and BOQs for specialized facilities.

Table 3:Kilifi County Adaptation Action and Implementation Plan

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
Thematic area 1: Water Security and Management				
Rainwater harvesting through Installation of gutters on house roofs and water tanks at household level in Mnarani (Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Wards	Rainwater harvesting through installation of gutters and high-capacity water tanks (10m^3) in schools– one school in each village per ward (Mnarani (6), Mwarakaya (4) and Sokoke (4)) to enhance water access by women and children	Drilling and solarizing boreholes in Mnarani (Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Wards	Kilifi County Government-FLLoCA, NGOs and Private Sector	50,000/HH, 400,000/School, 5,000,000/borehole
Fencing of springs in Mwarakaya Ward (Vwevwesi village)	Strategic (where embankment for water intake is viable) installation of solar powered water pumps on existing wells to reduce women workload (turn-around time) in Mnarani (Takaungu, Vijiweni and Vuma villages), Mwarakaya (Ondoni and Vwevwesi villages) and Sokoke (Midugani village)	Construction of water pans in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni and Vwevwesi villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani) Construction of a sand dam along River Mwabanda in Mwarakaya Ward (Ondoni village)		100,000 per spring (Fencing), 70,000/ water pump, 3,000,000/water pan, 1,000,000/sand dam
Sensitizing women and youth on water treatment in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala and Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele			KIMAWASCO, Kilifi County Government, CHPs, NGOs and Private Sector	Public barazas for the three wards at 280,000

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
and Ndugumnani villages) Wards				
Thematic area 2: Food Security, Ecosystem Restoration and Livelihoods				
Training and supporting women and youth on Resilience Design technique from homestead to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design - to maximize on time and use of local inputs to reduce farming costs)	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in Mnarani, Mwarakaya and Sokoke wards	Kilifi County Government, KARLO, NGOs and Private Sector	Farmer Field Schoos (FFFs) 10,000 per farmer for a 12-month cycle
Training women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as katumani maize cassava, sunflower, groundnuts, cashewnuts, sorghum), silage/hay making to address livestock feed challenges during drought in Mnarani (Midzimitsano, Mkwajuni, Takaungu,Timboni, Vijiweni, Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala, and Vwevwesi villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages)	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni, Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala, and Vwevwesi villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) to trained women and youth groups	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value addition and agribusiness opportunities Training women, youth and PADs on cassava value addition in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni, Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala, and Vwevwesi villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages)		Production costs: Maize 55,000 per acre and Cassava 40,000 per acre 10,000 for livestock farmers per year or crop cycle (Free if it is provided by the government)
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Mnarani (Midzimitsano, Takaungu, Timboni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala, and Vwevwesi villages) and			Public barazas for the three wards at 280,000, 55,000 per acre for pasture, 15,000 per Gala goat, 100,000 per dairy cow

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
	Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages			
Facilitate/train women and youth to establish tree nurseries of priority species (Annex 1) for on-farm tree production (Agroforestry, woodlots and boundary planting) in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala and Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Wards Prepare a Conservation and Management Plan for Kaya Chonyi and build the capacity of Kaya elders to support implementation in Mwarakaya Ward	Support MTAKIMAU CFA in Mnarani Ward to implement its PFMP programs Rehabilitate degraded areas in Sokoke Ward (Mwangea hills and Kaya Kadzistoni) Rehabilitate degraded areas in Mwarakaya Ward (Kaya Chonyi)		Kilifi County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per person per day for training 75,000 per nursery CFA support based on PFMP budget lines and priorities 150,000 per acre for rehabilitation
Sensitization and training women and youth on mangrove restoration (nursery establishment, transplanting, tending and monitoring) in Takaungu Creek, Mnarani Ward	Assessment and provision of technical support to the existing mangrove nurseries in Takaungu Creek, Mnarani Ward to ensure quality and sustainable seedlings supply for restoration interventions	Implement an integrated mangrove and coral reef restoration programme in Mnarani Ward;Takaungu Creek and Timboni village to enhance shoreline protection and ecosystem resilience respectively	KFS, KEFRI, Kilifi County Government, CFAs, NGOs, Private Sector	5,000 per person per day for training 75,000 per nursery 450,000/acre mangrove restoration, 3,000,000/acre coral reef restoration
Training women, youth and PADs on making organic briquettes (coconut husks,	Provision of energy saving jikos and kiln in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma			5,000 per person per day for training

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
cassava peels) in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala and Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Wards	villages), Mwarakaya (Mbuyuni, Ondoni, Silala and Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Wards			3,000 energy saving jiko & 7,000 per Kiln
Sensitizing women and youth on proper waste management practices (compositing organic waste, sorting, handling, disposal, upcycling) in Mnarani Ward (Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages) Sensitization on riverbanks management, flood control and sand harvesting impacts along Rivers Mwanzae and Mwabanda in Mwarakaya Ward Sensitization on climate change and environmental conservation in Sokoke Ward (Dzikunze, Midugani, Mwele and Ndugumnani villages)	Provision of central waste collection bins in shopping centres and linkage with waste collectors in Mnarani Ward (Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages) Regulation of harvesting sand along Rivers in Silala	Construct terraces and gabions to prevent soil erosion and manage runoff in Mnarani Ward (Vijiweni village) Construct gabions and check dams along River Mwanzae, Mwabanda and eroded sections to reduce flood velocity and stabilize banks in Mwarakaya Ward (Mbuyuni, Ondoni and Silala Villages)	NEMA, WRA, Kilifi County Government, NGOs, Private Sector	5,000 per person per day for training 100,000 for waste collection bin BOQ for terraces, gabions and check dams construction
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth and PADs on alternative livelihoods such as beekeeping and poultry	Provision of alternative livelihoods establishment materials/Equipment; modern beehives and egg incubators Mnarani	Construction of a cultural centre and establishment of a	KeFS, NMK, KMFRI Kilifi County Government, BMUs, NGOs, Private Sector	5,000 per person per day for training, 7,000-

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
farming in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala and Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Wards; ecotourism in Mnarani (mangrove boardwalk in Takaungu Creek), Mwarakaya (Kaya Chonyi and Mawe meru); mariculture (milk fish, mud crab) and aquaculture (catfish, tilapia) in Mnarani (Takaungu and Midzimitsano villages respectively)	(Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala and Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Wards; boards, fingerlings, cages, nets, boat, pond, and fish feed pellet mill (Midzimitsano and Takaungu villages) in in Mnarani (Takaungu and Midzimitsano villages) Provision of relief food and water to households during drought periods in Mnarani (Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Vwevwesi Villages) and Sokoke (Dzikunze, Mwele and Ndugumnani villages) Supporting women, youth and PADs in establishing small scale enterprises (stalls) at household level in Sokoke Ward (Dzikunze, Mwele and Ndugumnani villages)	nature trail in Kaya Chonyi to promote tourism		Modern beehive, 100,000-egg incubator, 1,000,000 boardwalk, 15 per fingerling, 1,000,000 cage, 30,000 net, 2,000,000 boat, 100,000 fish feed pellet mill and 1,000,000 pond, Relief food and water on a need basis, 50,000/HH for SMEs establishment
Promote basketry and mat weaving (already existing) through providing market linkage initiatives in Mnarani (Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages), Mwrakaya (Silala and Vwevwesi villages) – through exhibitions	Supporting women and youth to initiate and register a mangrove blue carbon project (Plan Vivo Systems and Standards) in Takaungu Creek, Mnarani Ward similar to Vanga blue carbon projects and consistent with the Climate Change (Carbon Markets) Regulations, 2024		NEMA, KEFRI, KFS, Kilifi County Government, CFAs, NGOs, Private Sector	200,000-exhibition 10,000 initial application fee, 100,000 project design document application fee, 1,000,000 feasibility study
Train VSLAs on Governance and alternative regenerative investment opportunities such as green bonds and money	Promote VSLAs excess savings investments in reputable interest earning schemes such as the green bond by Safaricom		County Government of Kilifi, NGOs, Private Sector	5,000 per person per day for training 100,000 per VSLA

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
market funds to diversify interest income from surplus savings				
Thematic area 4: Community Infrastructure and disaster management				
Train selected community emergency response teams in Mnarani (Midzimitsano, Timboni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni and Silala villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Undertake tree planting to address strong winds in Mnarani (Vuma village) and Sokoke (Ndugumnani village) Designate flood and wind disaster assembly areas and temporary camps for vulnerable communities in Mnarani Ward (Midzimitsano, Timboni and Vuma villages)	Design and implement a food for work programme during drought targeting the youth to restore drainage channels, clean culverts, construct gabions and plant trees for erosion and flood control	Construct raised footbridges/culverts at flood-prone crossing points in Mwarakaya Ward (Mbuyuni, Ondoni and Silala villages) Improve road drainage and install culverts to channel floodwater in Mnarani (Timboni village) and Mwarakaya (Ondoni village)	Kilifi County Government, KERRA, NGOs and Private Sector	BOQs required based on numbers and reputable training institutions
	Develop and operationalize Early Warning Systems (EWS) for floods, drought, strong winds, pests and diseases (forecasts via chiefs, barazas, media) in Mnarani (Midzimitsano, Takaungu, Timboni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala, and Vwevwesi villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages)	Equip community emergency response teams (PPEs, first aid kits) in Mnarani (Midzimitsano, Takaungu, Timboni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni and Silala villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages) Construct a health facility in Mnarani (Vijiweni and Vuma villages, Mwarakaya (Silala and Vwevwesi villages) Sokoke (Mwele village)	KMD, NDMA Kilifi County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	5,000 per day per trainee 840,000 for EWS in the three wards 200,000 for disaster assembly area and temporary camp BOQ for equipment and health facility
Thematic area 5: Conflicts resolution (Human Wildlife Conflicts)				
Sensitize Mwarakaya (Ondoni and Vwevwesi villages) and Sokoke (Dzikunze, Mwele,	Promote chili cultivation (provide training and quality seeds) and apiculture (provide training and hives) along forest–farmland	Installation of solar power electric fence around Kaya Chonyi in Mwarakaya Ward	KWS, Kilifi County Government, NGOs, Private sector	Public barazas for the two wards at 120,000, 350/packet-chilli,

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
Midugani and Ndugumnani villages) wards communities on potential conflict hotspots, minimizing human-wildlife interactions and reporting to KWS (hotline) as well as compensation procedures	boundaries as a Nature-Based Solution (NBS) in Sokoke ward (Dzikunze, Midugani, Mwele and Ndugumnani villages)			5,000,000- Death 3,000,000- Permanent Disability 2,000,000- Any other injury Crop/Property destruction- compensation based on market rates, 7,000,000 for electric fence
Thematic area 6: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought, floods and strong winds climate hazards	Develop and implement a Ward GBV anticipatory action plan in collaboration with relevant partners	Establish GBV recovery center (this could be at the county level)	National Government, Kilifi County Government and NGOs, National Government, Kilifi County Government, National Police Service, NGOs, GVRC, COVAW, HAK	1,000,000 for consultancy services Construction of GBV recovery center subject to a detailed Bill of Quantities
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV anticipatory actions ahead of climate hazards	Strengthen gender-responsive governance-inclusion of women, youth and PADs through capacity building in Mnarani (Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages), Mwarakaya (Mbuyuni, Ondoni, Silala and Vwevwesi Villages) and Sokoke (Dzikunze, Midugani, Mwele and Ndugumnani villages)		Kilifi County Government and NGOs	5,000 per person per day for training
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kilifi County Government, Civil Society, NGOs, Private Sector	5,000 per person per day for training BOQ for GBV response mechanisms

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
men in peaceful conflict resolution				
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kilifi County Government, NGOs, Private Sector	100,000- Boot camp, Vaccination at local medical facilities, BOQ required for drug rehabilitation centre School feeding and scholarships needs assessment
Facilitate PADs to link with organizations that can provide infrastructural assistance such as wheelchairs, specialized training and entrepreneur skills			Kilifi County Government, NGOs, Private Sector	50,000 for meetings

This implementation matrix will guide the roll-out of adaptation projects in a coordinated manner. Each action will be incorporated into annual work plans of the relevant county departments and into Ward Adaptation Plans for community execution. The responsible entities listed will form implementation teams or sub-committees as needed (e.g. a Water Project Team for boreholes, an Agro-Pastoral Team for farm interventions) to manage day-to-day activities. The Ward Climate Change Planning Committees will monitor progress on the ground and report any issues or delays to the County Climate Change Unit through the quarterly reports.

Several actions (such as water infrastructure, bridges, and coastal defenses) require technical design and feasibility assessments upfront. The county will leverage engineering staff and where needed external experts to conduct these studies in 2025–2026 so that projects are well-designed and meet standards (including environmental impact assessments for bigger projects, overseen by NEMA). Procurement of contractors for works will follow public procurement laws, with a preference for labor-intensive approaches that generate local employment (e.g. engaging youth/women groups in nursery operations, tree planting, and routine maintenance). For community-level actions, funds channeled via the CCCF or grants will be managed through community structures with transparency and accountability measures (WCCPC oversight committees will track use of funds and assets).

5.3 Financing and resource mobilization

Implementing the plan will require a combination of county budget allocation, climate funds, national programs, and partner support. The Kilifi County Government is expected to allocate a portion of its development budget each year to priority adaptation projects (e.g. borehole drilling in Water Dept budget, irrigation schemes in Agriculture Dept, emergency funds in Disaster unit). As mandated by the Climate Change Act 2023, all county sectors must allocate at least 2% of their budget to climate change activities, which could translate to roughly KES 2–3 million per year per ward for adaptation if secured. The County Climate Change Fund (CCCF) is a dedicated source that will finance many community-prioritized actions; the WCCPCs will prepare proposals to tap the CCCF regularly. Over five years, an estimated KES 10 million (aggregate) could be drawn from the CCCF for ward-level projects in Kilifi if well-coordinated.

National government programs provide additional avenues: the plan will align projects to initiatives like the National Drought Emergency Fund (for water projects and drought mitigation), the Kenya Climate Smart Agriculture Project (KCSAP) (for promoting resilient crops and farming practices), and the Kenya Livestock Insurance Program (KLIP) (for livestock insurance). Where applicable, the county will seek funding from the National Climate Change Fund (NCCF) once operational and continue engaging with the National Treasury's Climate Finance Unit to access any performance-based grants under FLLoCA.

Development partners and non-governmental organizations (NGOs) currently operating in Kilifi County which include World Bank, World Wildlife Fund (WWF), Childcare Worldwide, Plan International, the Kenya Cereal Enhancement Programme (KCEP), Botanical Garden Conservation International (BGCI), and Anglican Development Services, Pwani (ADS-Pwani)are anticipated to provide critical co-financing and technical assistance in the implementation of these adaptation actions. Ongoing initiatives (for example: FLLoCA and the Restore Africa Program) will be strategically aligned with the County Climate Adaptation Action Plan (CCAAP) to ensure coherence, leverage resources, and optimize impact. These partners are expected to support capacity development, community awareness, and the provision of climate-resilient assets, (for example: tanks, farm inputs and seedlings), to complement county-level investments. International development financiers such as the German Development Bank (KfW), the United Nations Development Programme (UNDP), the Global Environment Facility (GEF), and the

Green Climate Fund (GCF) may channel climate finance through national mechanisms or accredited implementing partners, such as NGOs and civil society organizations, to strengthen adaptive capacity at local levels. The private sector will form a strategic partner in mobilizing additional resources and fostering innovation through public–private partnerships and corporate social responsibility (CSR) frameworks. For example, telecommunications providers could enhance Early Warning Systems through SMS dissemination.

The implementation will follow a flexible, learning-by-doing approach: annual work plans will be reviewed and updated based on progress, new climate information, and community feedback. The County Climate Change Unit, working with WCCPCs, will produce an annual implementation report that details what was done, funds used, and results achieved for each action. This report will allow the County Adaptation Planning Committee (as established by the County Climate Change Act) to re-prioritize or add new actions if needed, ensuring the plan remains dynamic and responsive through 2030.

6 Chapter 6: Monitoring, Evaluation, Reporting, and Learning (MERL) Framework

6.1 Introduction

The Monitoring, Evaluation, Reporting, and Learning (MERL) Framework for the County Climate Change Adaptation Plan (CCCAP) provides a comprehensive structure for tracking progress, assessing effectiveness and enhancing accountability in the implementation of climate adaptation initiatives. It ensures that all activities undertaken across the county are evidence-based, results-oriented and aligned with both national and county development priorities. The framework is anchored within the institutional arrangements established under the County Climate Change Unit (CCCU), which serves as the coordinating body responsible for guiding implementation, consolidating data, and overseeing reporting on adaptation interventions. Working closely with the County Adaptation Planning Committee (CAPC) and the Ward Climate Change Planning Committees (WCCPCs), the CCCU facilitates participatory monitoring and learning processes that capture community perspectives, local innovations, and lessons from the ground.

This framework is guided by the provisions of the Climate Change Act (2023), which mandates both national and county governments to integrate climate change actions into their development planning, budgeting and implementation processes. It is also closely aligned with national and county policy instruments, including the National Adaptation Plan (NAP), which provides a strategic direction for climate adaptation across sectors, and the National Climate Change Action Plan (NCCAP), which outlines priority actions for achieving Kenya's climate resilience and low-carbon development goals. At the county level, the framework integrates with the County Integrated Development Plan (CIDP) to ensure coherence between climate change adaptation and the broader development agenda. By establishing clear indicators, data management systems, and feedback mechanisms, the MERL framework enables systematic learning, adaptive management, and transparent reporting to stakeholders. Ultimately, it strengthens institutional capacity, promotes accountability, and fosters continuous improvement in building climate resilience and sustainable livelihoods across Kilifi County.

6.2 The County Theory of Change

The County's Theory of Change (ToC) outlines the logical pathways through which adaptation inputs and activities translate into outputs, outcomes, and long-term resilience impacts. It articulates how county-level investments in climate-resilient infrastructure, capacity building, institutional strengthening, and sustainable livelihoods collectively reduce vulnerability to climate hazards.

- **Inputs/Activities:** Financial allocations from the County Climate Change Fund (CCCF), national transfers, development partner support, and technical assistance will finance activities such as construction of resilient water infrastructure, promotion of climate-smart agriculture, enforcement of environmental bylaws, and establishment of ecosystem restoration programs.

- **Outputs:** Tangible products from these activities include functional water systems and irrigation infrastructure, trained farmers and pastoralists adopting improved practices, rehabilitated forest and mangrove ecosystems, enhanced early warning dissemination, and operational local climate institutions across all sub-counties.

- **Outcomes:** The expected short- to medium-term effects include improved access to safe water during drought periods, stabilized agricultural yields, diversified income sources, strengthened institutional coordination, and enhanced preparedness and response to climate hazards across the county.

- **Impact (Goal):** The overarching goal is to enhance the adaptive capacity, resilience, and socio-economic well-being of communities in the county. This means reduced exposure and sensitivity to climate hazards, strengthened ecosystem services, improved food and water security, and sustained livelihoods despite climate variability.

The ToC assumes enabling conditions such as consistent policy support, effective inter-departmental collaboration, and active participation of marginalized groups including women, youth, and Persons Abled Differently (PAD). Gender and social inclusion remain central: by empowering women and youth in decision-making and livelihood diversification, the county enhances equitable resilience-building and intergenerational adaptation capacity.

6.3 Performance Indicators for Adaptation

The County's MEL system is built around performance indicators structured along the ToC levels; inputs, outputs, outcomes, and impacts. These indicators are harmonized with Kenya's National Adaptation Plan (NAP) and National Climate Change Action Plan (NCCAP) indicators, ensuring consistency in national reporting while responding to county-specific adaptation priorities.

Input/Output Indicators:

1. WATER SECURITY AND MANAGEMENT

- Number of household rainwater harvesting systems installed (gutters + tanks).
- Number of schools equipped with high-capacity rainwater harvesting systems.
- Number of water pans constructed or desilted.
- Length/coverage of pipelines and kiosks constructed (e.g., Mkwiro–Wasini).
- Number of manual and solar-powered pumps installed on existing wells.
- Number of women, youth, and PADs using solarised/manual hand pumps.
- Number and percentage of women, youth, and PADs in water management committees.
- Number of village water storage structures constructed (Majibao, elevated tanks).
- Amount of budget allocated and utilised for water infrastructure.
- Number of boreholes drilled, equipped, and solarised.
- Number of boreholes fully operational and supplying water year-round.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Number of farmers trained on Resilience Design techniques.
- Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
- Acreage under drought-tolerant crops supported with certified seeds.
- Number of Farmer Field Schools established and functional.
- Number of demonstration farms established by women and youth lead farmers.
- Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
- Number of households supported with organic manure production and composting.
- Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
- Number of livestock fodder seed banks and pasture plots established.
- Number of households receiving improved livestock breeds (Gala goats, dairy cows).
- Number of farmers sensitised on crop/livestock insurance.
- Number of households supported with grain storage equipment (hermetic bags, grain banks).
- Number of village pest and disease surveillance teams established and trained.

- Number of households receiving indigenous/fruit seedlings.
- Number of community seedling distribution events held.
- Number of on-farm tree planting sessions conducted.
- Number of mangrove nurseries supported and maintained.
- Number of acres of mangrove restored through enrichment planting.
- Number of community trainings on mangrove protection and sustainable harvesting.
- Number of NBS windbreaks (Neem, Casuarina) planted across all villages.
- Number of environmental conservation awareness sessions held.
- Number of households adopting sustainable charcoal alternatives or clean energy.
- Budget allocated and utilised for restoration activities.

3.DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Number of women and youth trained in alternative livelihoods (eco-tourism, beekeeping, poultry).
- Number of modern beehives, incubators, or eco-tourism boats distributed.
- Number of households supported with livelihood start-up kits.
- Number of community groups formed for value addition and market linkages.
- Number of training sessions on business development, value addition, and financial literacy.
- Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
- Number of school feeding programmes supported.
- Budget allocations for livelihood diversification implemented.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Number of culverts reconstructed or improved with proper drainage.
- Number of flood-prone crossing points fitted with raised footbridges.
- Number of youth mobilised in food-for-work programmes for drainage, gabions, and erosion control.
- Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.
- Number of community dissemination channels established (chief barazas, noticeboards, SMS).
- Number of community emergency response teams trained.
- Number of community emergency response teams equipped (boats, PPE, first-aid kits).
- Number of designated disaster assembly points established and labelled.
- Number of community trainings on resilient building practices conducted.
- Number of households implementing safer roofing techniques (replacing strings with nails).
- Budget allocated and utilised for infrastructure upgrades (roads, culverts, drainage).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Number of community sensitization sessions on human–wildlife conflict hotspots.
- Number of households supported with HWC reporting mechanisms (hotline awareness).
- Number of kilometres of solar-powered fencing installed in hotspot areas (e.g., Kiwambale).
- Number of community trainings on safe behaviour in crocodile, baboon, and snake-risk zones.
- Number of reported HWC cases logged and submitted to KWS for follow-up.

- Funding allocated and utilised for HWC mitigation (fencing, awareness, surveillance).

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Number of gender-responsive governance trainings conducted.
- Number and percentage of women, youth, and PADs participating in ward committees.
- Number of VSLAs supported with GBV awareness and rights training.
- Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
- Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
- Number of early warning systems integrating GBV alert messaging.
- Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.
- Number of youth bootcamps conducted on drugs, behaviour change, and reproductive health.
- Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
- Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT

- Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
- Increase in household water storage capacity (litres/HH).
- Percentage of water facilities functional year-round.
- Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).
- Increased reliability of water supply during drought periods (days/months of service continuity).

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Stability of crop yields over 3–5 seasons among supported farmers.
- Improvement in soil health indicators, including:
- Increase in soil organic matter (%)
- Increase in soil moisture retention capacity
- Improvement in soil nutrient status (N, P, K)
- Improved aggregate stability and soil structure
- Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
- Increase in acreage under diversified and climate-resilient farm systems.
- Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
- Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles.
- Increase in on-farm and community tree cover (%) over baseline.
- Increase in mangrove seedling survival rates and total acreage restored.
- Reduction in incidents of illegal harvesting, charcoal burning, and mangrove cutting.
- Increase in households adopting windbreak and shelterbelt systems.
- Increased community participation in environmental conservation and sustainable harvesting practices.

3.DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping, seaweed value addition).
- Increase in the number of households with multiple income streams across seasons.
- Percentage of women, youth, and PADs accessing new livelihood opportunities.
- Increased financial literacy and business skills among trained groups.
- Increase in households benefiting from school feeding, cash transfers, or food-for-work schemes.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Reduction in households exposed to flood-prone or erosion hotspots.
- Percentage of villages receiving timely Early Warning System (EWS) alerts.
- Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
- Reduction in flood-related damages to dwellings and access roads.
- Level of preparedness and readiness of community emergency response teams (measured via drills).
- Increased adoption of wind-resilient building practices in households.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Reduction in frequency of human–wildlife conflict incidents (crocodile attacks, baboon raids).
- Increase in the number of households protected by solar-powered fences.
- Increase in community capacity to report and respond to HWC events (via KWS hotline).
- Increased community compliance with safe-behaviour protocols in HWC hotspots.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.
- Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
- Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
- Increase in women's ownership of productive assets (livestock, seaweed farms, equipment).
- Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
- Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT

- Percentage of the ward population with year-round access to safe and reliable water.
- Increase in average litres of water available per person per day across villages.
- Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.

- Sustained decrease in waterborne diseases due to improved water quality and availability.
- Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Sustained increase in crop yields above baseline across 3–5 year periods.
- Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
- Measurable improvements in soil health at landscape level, including:
- Higher soil organic carbon (SOC)
- Increased vegetation cover (NDVI index)
- Reduced erosion and sediment loss
- Enhanced household food security across all seasons.
- Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).
- Increases in forest cover, on-farm trees, and mangrove extent compared to baseline.
- Restoration of degraded landscapes evidenced by improvements in NDVI and tree density.
- Reduction in illegal harvesting, mangrove cutting, and charcoal burning.
- Enhanced ecosystem services, including shoreline protection, soil stability, and microclimate regulation.
- Long-term improvement in habitat quality for biodiversity and fisheries productivity.

3.DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, seaweed, poultry).
- Improved household resilience to income shocks from climate hazards.
- Decreased poverty levels due to stable and multi-stream livelihoods.
- Increased school attendance and nutritional outcomes for children due to social protection schemes.
- Greater financial stability among women, youth, and PAD households.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Significant reduction in climate-related damage to homes, roads, and community facilities.
- Communities demonstrating effective disaster preparedness and response during floods and storms.
- Long-term reduction in displacement, injuries, and asset losses during climate hazards.
- Increased structural safety and durability of settlements in strong-wind and flood-prone areas.

5.CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Long-term reduction in human injuries, livestock losses, and crop destruction from wildlife.
- Improved coexistence between communities and wildlife due to effective mitigation (fencing, reporting systems).

- Reduced economic losses from HWC-related disruptions.
- Increased safety and mobility for women, children, and PADs in high-risk zones.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.
- Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
- Increased leadership roles held by women, youth, and PADs in adaptation implementation.
- Growth in the number of marginalized and vulnerable groups leading or influencing climate adaptation actions.
- Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory action systems.

Baseline data will be established using county-level assessments, socio-economic surveys, and historical climate datasets. Data will be disaggregated by gender, livelihood zone, and administrative unit to assess differential impacts and ensure inclusive adaptation outcomes. The County Climate Change Unit (CCCU) will coordinate data collection and analysis through a centralized digital monitoring system linked to national reporting platforms.

6.4 Adaptation Reporting and Learning

Reporting: The county will adopt a tiered reporting structure integrating ward, sub-county, and county levels. WCCPCs will compile quarterly progress updates feeding into sub-county adaptation reports, which the CCCU will synthesize into a county-level Annual Climate Adaptation Progress Report. This report will highlight achievements against indicators, emerging challenges, expenditure tracking, and gender-disaggregated performance metrics. It will also inform the County Annual Development Plan (ADP) and the State of the Environment (SoE) reporting.

Learning and Reflective Review: Mid-term (Year 3) and end-term (Year 5) evaluations will be conducted to assess performance, effectiveness, and sustainability of interventions. Participatory learning workshops will bring together county departments, communities, development partners, and academia to review progress, identify lessons, and revise priorities. Tools such as outcome harvesting, community scorecards, and Most Significant Change (MSC) narratives will be applied to capture qualitative insights. Findings will guide evidence-based revisions of the CCAAP and inform the subsequent planning cycles.

Knowledge Sharing and feedback to Policy: The county will institutionalize knowledge management through a digital climate knowledge hub, documenting case studies, research findings, and best practices. Cross-county exchange visits and inter-ward learning forums will promote horizontal learning, while upward policy feedback loops will relay local adaptation evidence to national institutions such as the National Climate Change Directorate (NCCD) and National Drought Management Authority (NDMA). This ensures county-level insights inform national adaptation strategies, fulfilling obligations under Section 25 of the Climate Change Act (2023).

Sustainability of M&E: To ensure continuity, the county will designate a Climate M&E Focal Officer within the CCCU to oversee data management, capacity building, and integration of adaptation indicators into the County Integrated Monitoring and Evaluation System (CIMES).

MEL responsibilities will also be mainstreamed into departmental workplans, ensuring adaptation tracking becomes part of routine service delivery.

7 Chapter 7: Risk-Informed Development and Disaster Management Integration

Kilifi's CCAAP takes a risk-informed development approach, meaning it integrates climate and disaster risk considerations into all development planning and ensures coordination with disaster management structures. This approach recognizes that adaptation and disaster risk reduction (DRR) are two sides of the same coin: long-term resilience-building must go hand in hand with preparedness for extreme events. Several mechanisms are put in place to achieve this integration:

Mainstreaming Disaster Risk Reduction into Adaptation Actions: Each adaptation intervention has been designed with disaster scenarios in mind. For instance, the water security actions (boreholes, dams) are not only for gradual climate change adaptation but also function as drought emergency buffers when rains fail. Similarly, strengthening houses and early warning systems for storms is part of adaptation and directly reduces disaster impacts. The plan aligns with the NAP 2015–2030 directive to embed early warning systems and disaster preparedness into development. By doing so, development gains (like new infrastructure or agricultural productivity improvements) are protected from being wiped out by the next flood or cyclone.

Unified Local Committees for Climate and Disaster Management: At the ward level, the institutional arrangement deliberately overlaps the adaptation committee (WCCPC) with the Ward Disaster Risk Management Committee. In many wards, rather than having separate groups, the WCCPC either serves as the disaster committee or has a significant membership overlap (with additional co-opted members like local Red Cross volunteers or disaster response teams). This ensures seamless transition between "normal times" (where the focus is on resilience projects) and "disaster times" (where focus shifts to emergency response). For example, the same local committee that implements flood control measures also spearheads evacuations or relief distribution during an actual flood. Clear protocols will define roles: when a drought is looming (as signaled by NDMA early warning bulletins), the WCCPC/disaster committee will activate contingency plans such as water trucking or food aid linked with the national safety net programs. The integration avoids duplication of structures and information gaps, the people planning adaptation are the same who manage immediate response, leading to more coherent risk management.

Climate Information and Early Warning Systems: A core component of the plan is enhancing early warning systems (EWS) for climate hazards as a bridge between adaptation and DRR. The county, through KMD and NDMA, will ensure that seasonal climate forecasts, drought early warning stage updates, and extreme weather alerts are rapidly communicated to communities in user-friendly formats. The plan supports training of local leaders and volunteers on how to interpret these advisories and take early actions (e.g. plant early-maturing crops if rains are predicted to delay, pre-position animal feed if a drought warning is issued, or evacuate to safe ground when heavy rainfall/flood alerts come). Every year before the rainy seasons, the WCCPCs and County Disaster Management team will hold pre-season planning meetings to review the weather outlook and agree on preparedness measures (cleaning drains, alerting farmers, updating stockpiles of relief items). This operationalizes the concept of anticipatory action, which is advocated in Kenya's DRM policy and the NAP. For sudden-onset events, the integration with disaster response means that as soon as, say, a flood hits, the adaptation structures can document losses and start planning build-back-better recovery (like rebuilding houses with improved designs).

Development Planning with Risk Lens: The County government will use climate and hazard risk data to inform where and how development projects are implemented. For example, infrastructure projects in CIDP (roads, schools, water systems) will undergo climate risk screening, if a proposed road goes through a floodplain, the design will include sufficient

drainage and elevation (or be rerouted if too high-risk); new water supply schemes will consider future rainfall variability in sizing storage. Settlements expansion plans near the coast will incorporate sea-level rise projections to avoid placing critical facilities in zones likely to be inundated by 2050. This risk-informed planning is supported by the vulnerability maps and data from the ward assessments. The County Department of Planning, in concert with the Climate Change Unit and NEMA, will ensure Environmental Impact Assessments (EIAs) for major projects explicitly address climate and disaster risks, as now required by regulations. Through these steps, climate-proofing development becomes a standard practice rather than an afterthought.

Linkages with Disaster Response Mechanisms: The CCAAP explicitly links with existing disaster response funds and protocols. The National Drought Emergency Fund (NDEF) and other rapid response funds (e.g. Kenyans for Kenya, Kenya Red Cross emergency funds) will be activated as per NDMA's drought phase classification to support actions like water trucking, food relief, or destocking during severe drought, these are written into the drought contingency plans at ward level. The plan's MERL will track not just adaptation indicators but also disaster loss data (people affected, economic losses from events) to gauge if adaptation efforts are reducing these over time. Furthermore, the adaptation projects are designed to have dual use: for instance, rainwater harvesting tanks installed as adaptation also serve as emergency water sources in a sudden water crisis; grain banks set up to preserve surplus can double as food distribution hubs in a famine, etc. The presence of trained community responders (through Red Cross and St. John Ambulance collaborations) means that when a disaster strikes, local capacity is there to act while awaiting external help. Overall, the plan fosters a culture of prevention and preparedness in line with the Sendai Framework for DRR, by building local systems that anticipate hazards and minimize impact, rather than purely reacting afterward.

Continuous Risk Assessment and Scenario Planning: The County Climate Change Unit, in partnership with NDMA and KMD, will annually update a Climate Risk Register for the county, essentially a living document that identifies emerging risks (e.g. new disease vectors, potential landslide zones due to land use changes, etc.) and flags any adjustment needed in the adaptation plan. They will conduct scenario exercises, for example, "what if the next short rains fail completely?" or "what if a cyclone makes landfall near Kilifi?" and use those to stress-test the readiness measures. This kind of forward-looking approach ensures that the adaptation actions remain relevant under a range of future conditions (including extreme worst-case events). The learning from actual disaster events will also feed into adaptation: if a flood causes damage in a new area, that area will be targeted for future flood control investment.

The disaster risk management is interwoven through the CCAAP, creating a resilient development pathway for Kilifi. By having joint committees, shared early warning systems, and risk considerations in development, the county will be able to protect development gains, reduce losses from climate extremes, and bounce back faster when events occur. This integrated approach is in full alignment with national priorities on Disaster Risk Management integration (as highlighted in NCCAP and the National Disaster Risk Management Policy) and positions KIlifi as a model for local climate-resilient development planning.

8 References

1. Mnarani Ward Adaptation Plan (2025) – Participatory Climate Risk and Vulnerability Assessment Findings and Adaptation Actions.
2. Mwarakaya Ward Adaptation Plan (2025) – Participatory Climate Risk and Vulnerability Assessment Findings and Adaptation Actions.
3. Sokoke Ward Adaptation Plan (2025) – Participatory Climate Risk and Vulnerability Assessment Findings and Adaptation Actions.
4. Kenya National Adaptation Plan (2015–2030) – National Climate Change Priorities and Mainstreaming Guidelines.
5. National Climate Change Action Plan (NCCAP) 2023–2027 – Priority Adaptation Actions and Financing Mechanisms (FLLoCA, etc.).
6. Kilifi County Climate Change Action Plan (2023–2027) – County Climate Change Act 2021 provisions, County Climate Change Fund, and Sectoral Targets.
7. Kilifi County Integrated Development Plan (CIDP) 2023–2027 – Development Programs Mainstreaming Climate Resilience and Disaster Risk Management. (Kilifi County Planning Unit, 2022).
8. Participatory Climate Vulnerability Assessment Reports (2025) – Community hazard timelines, vulnerability scores, and adaptation proposals in Mnarani, Mwarakaya and Sokoke Wards
9. County Climate Change Act (Kilifi County, 2021) – Establishment of Climate Change Unit, Ward Committees, and Integration of Climate Change in County Functions.
10. Kenya Climate Change Act (2023) – Devolved Climate Finance and Reporting Requirements for Counties.

9 Acknowledgements

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